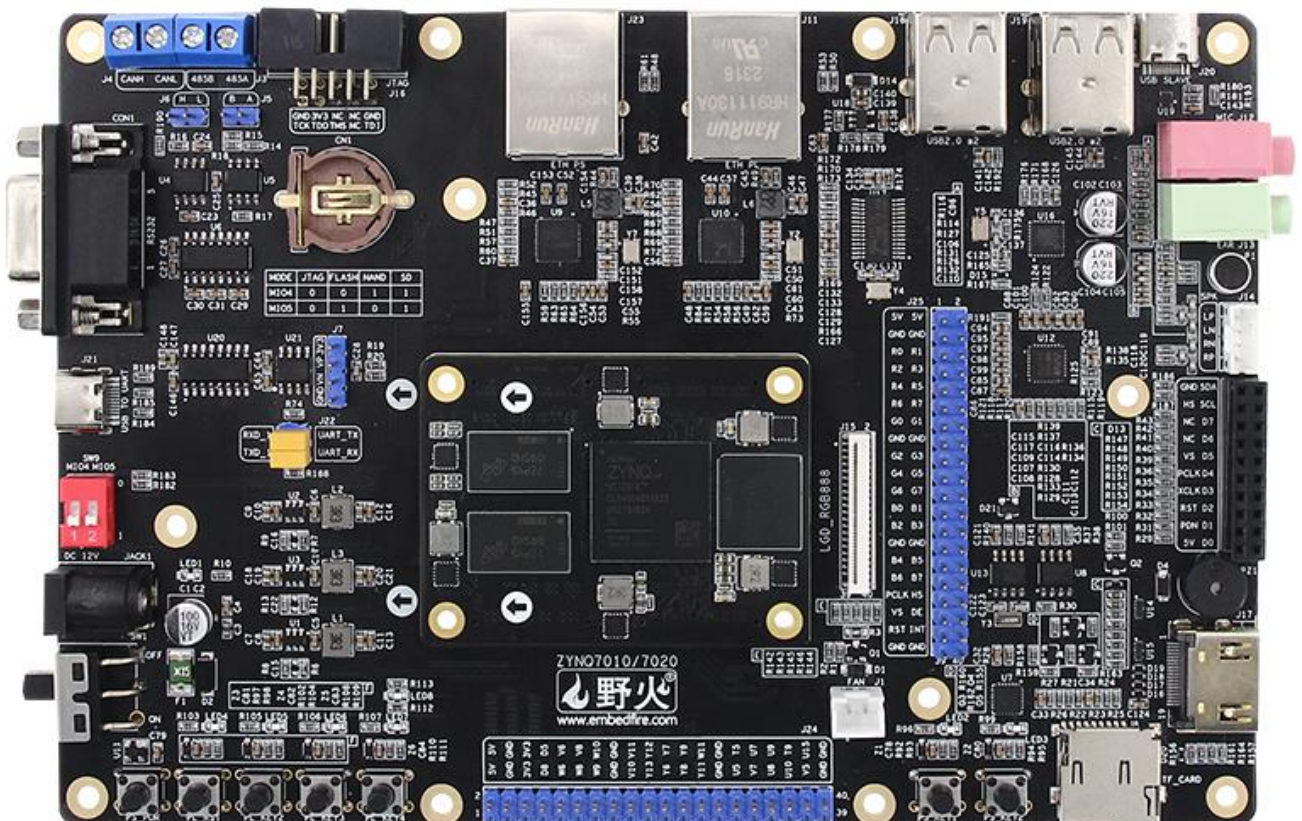


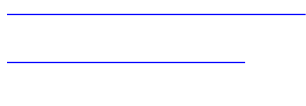
2025/01



Sales and Service Contact

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manual version

手册版本	日期	更新说明
V 1.0	2025-01	• 初始版本

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Chapter 1 Haoyue_Introduction to development board ch

We have two different core boards. The difference between these two core boards lies in the difference in their FPGA main chips. Therefore, the difference between the two core boards lies in the difference in FPGA chips.

The two FPGA chips we use are Xilinx's zynq7000 series chips; the specific models are: XC7Z010-1CLG400I, XC7Z020-2CLG400I.

The specific resource amounts of these two FPGA chips can be seen in the figure below (the figure is taken from the Xilinx

Feature Summary

Table 1: Zynq-7000 and Zynq-7000S SoCs

	Device Name	Z-7007S	Z-7012S	Z-7014S	Z-7010	Z-7015	Z-7020	Z-7030	Z-7035	Z-7045	Z-7100
	Part Number	XC7Z007S	XC7Z012S	XC7Z014S	XC7Z010	XC7Z015	XC7Z020	XC7Z030	XC7Z035	XC7Z045	XC7Z100
Processing System	Processor Core	Single-core ARM Cortex-A9 MPCore™ with CoreSight™			Dual-core ARM Cortex-A9 MPCore™ with CoreSight™						
	Processor Extensions	NEON™ & Single / Double Precision Floating Point for each processor									
	Maximum Frequency	667 MHz (-1); 766 MHz (-2)			667 MHz (-1); 766 MHz (-2); 866 MHz (-3)			667 MHz (-1); 800 MHz (-2); 1 GHz (-3)			667 MHz (-1) 800 MHz (-2)
	L1 Cache	32 KB Instruction, 32 KB data per processor									
	L2 Cache	512 KB									
	On-Chip Memory	256 KB									
	External Memory Support ⁽¹⁾	DDR3, DDR3L, DDR2, LPDDR2									
	External Static Memory Support ⁽¹⁾	2x Quad-SPI, NAND, NOR									
	DMA Channels	8 (4 dedicated to Programmable Logic)									
	Peripherals ⁽¹⁾	2x UART, 2x CAN 2.0B, 2x I2C, 2x SPI, 4x 32b GPIO									
	Peripherals w/ built-in DMA ⁽¹⁾	2x USB 2.0 (OTG), 2x Tri-mode Gigabit Ethernet, 2x SD/SDIO									
	Security ⁽²⁾	RSA Authentication, and AES and SHA 256-bit Decryption and Authentication for Secure Boot									
Processing System to Programmable Logic Interface Ports (Primary Interfaces & Interrupts Only)	2x AXI 32b Master 2x AXI 32-bit Slave 4x AXI 64-bit/32-bit Memory AXI 64-bit ACP 16 Interrupts										

Figure 1 zynq7000 series resources

Table 1: Zynq-7000 and Zynq-7000S SoCs (Cont'd)

	Device Name	Z-7007S	Z-7012S	Z-7014S	Z-7010	Z-7015	Z-7020	Z-7030	Z-7035	Z-7045	Z-7100
	Part Number	XC7Z007S	XC7Z012S	XC7Z014S	XC7Z010	XC7Z015	XC7Z020	XC7Z030	XC7Z035	XC7Z045	XC7Z100
Programmable Logic	Xilinx 7 Series Programmable Logic Equivalent	Artix®-7 FPGA	Artix-7 FPGA	Artix-7 FPGA	Artix-7 FPGA	Artix-7 FPGA	Artix-7 FPGA	Kintex®-7 FPGA	Kintex-7 FPGA	Kintex-7 FPGA	Kintex-7 FPGA
	Programmable Logic Cells	23K	55K	65K	28K	74K	85K	125K	275K	350K	444K
	Look-Up Tables (LUTs)	14,400	34,400	40,600	17,600	46,200	53,200	78,600	171,900	218,600	277,400
	Flip-Flops	28,800	68,800	81,200	35,200	92,400	106,400	157,200	343,800	437,200	554,800
	Block RAM (# 36 Kb Blocks)	1.8 Mb (50)	2.5 Mb (72)	3.8 Mb (107)	2.1 Mb (60)	3.3 Mb (95)	4.9 Mb (140)	9.3 Mb (265)	17.6 Mb (500)	19.2 Mb (545)	26.5 Mb (755)
	DSP Slices (18x25 MACCs)	66	120	170	80	160	220	400	900	900	2,020
	Peak DSP Performance (Symmetric FIR)	73 GMACs	131 GMACs	187 GMACs	100 GMACs	200 GMACs	276 GMACs	593 GMACs	1,334 GMACs	1,334 GMACs	2,622 GMACs
	PCI Express (Root Complex or Endpoint) ⁽³⁾		Gen2 x4			Gen2 x4		Gen2 x4	Gen2 x8	Gen2 x8	Gen2 x8
	Analog Mixed Signal (AMS) / XADC	2x 12 bit, MSPS ADCs with up to 17 Differential Inputs									
	Security ⁽²⁾	AES and SHA 256b for Boot Code and Programmable Logic Configuration, Decryption, and Authentication									

Notes:

1. Restrictions apply for CLG225 package. Refer to the [UG585, Zynq-7000 SoC Technical Reference Manual \(TRM\)](#) for details.
2. Security is shared by the Processing System and the Programmable Logic.
3. Refer to [PG054, 7 Series FPGAs Integrated Block for PCI Express](#) for PCI Express support in specific devices.

Figure 2 zynq7000 series resources (continued)

According to the above figure, you can clearly see the resource situation of each model of the zynq7000 series. Here we take XC7Z010-1CLG400I as an example. The main parameters are as follows:

Logic cells Logic cells: 28K Processor core: Dual-core ARM Cortex-A9 MPCore, maximum frequency 667MHZ DSP Slices: 80 LUT: 17600 Flip-flops: 35200 Block RAM: 2.1Mb

FPGA 芯片	皓月 XC7Z010-1CLG400I	皓月 XC7Z020-2CLG400I
封装	CLG400	CLG400
逻辑单元 Logic cells	28K	85K
处理器内核	双核 ARM Cortex-A9 MPCore, 最大频率 667MHZ	双核 ARM Cortex-A9 MPCore, 最大频率 766MHZ
DSP Slices	80	220
LUT	17600	53200
触发器	35200	106400
Block RAM	2.1Mb	4.9Mb
时钟单元 CMT	2	4
速度等级	-1	-2
工作温度	- 25°C ~ 85°C	- 25°C ~ 85°C
PCIe	无	无
高速收发器	无	无
PL 端 IO 口	99	124
PS 端 IO 口	42	

The physical pictures of the main chips of the two core boards are shown below:



Figure 2 Physical pictures of two FPGA chips

Chapter 2 Haoyue_Development Board Core Board Introduction

2.1 Core board appearance diagram

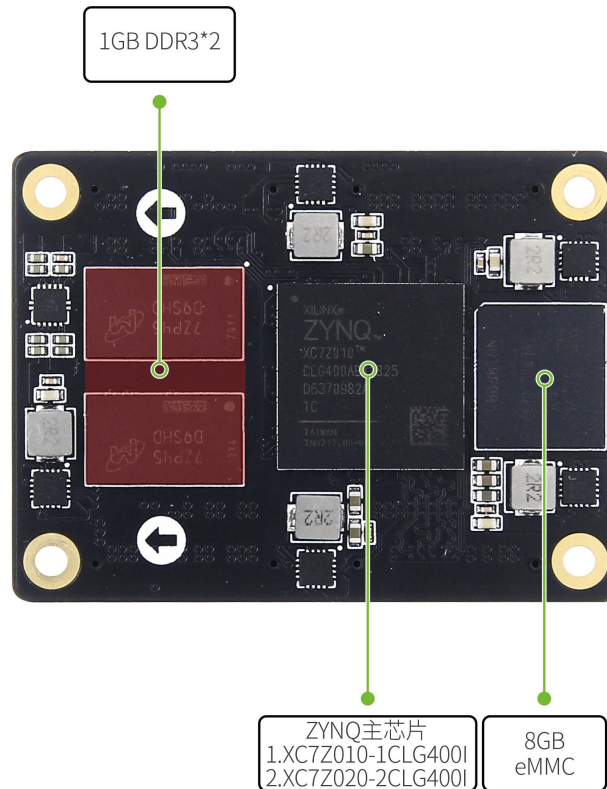


Figure 2.1-1 Haoyue_development board core board front view

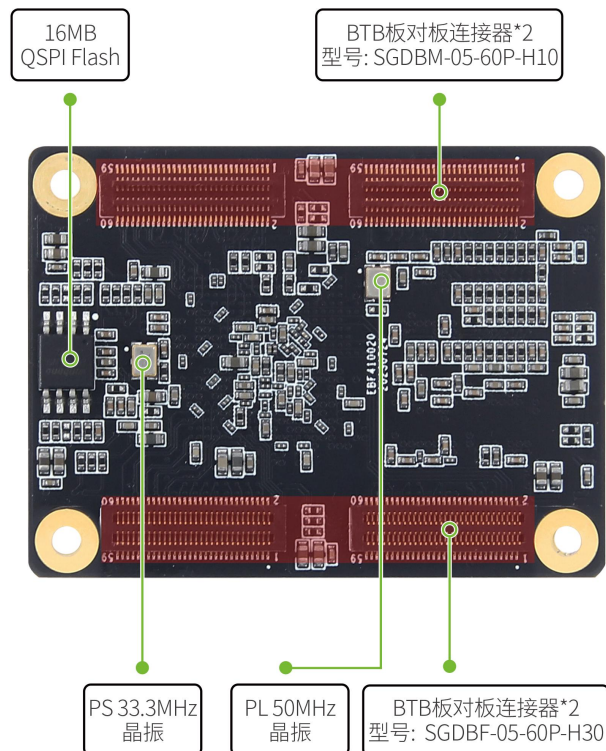


Figure 2.1-2 Haoyue_development board core board back view

2.2 Core board size diagram

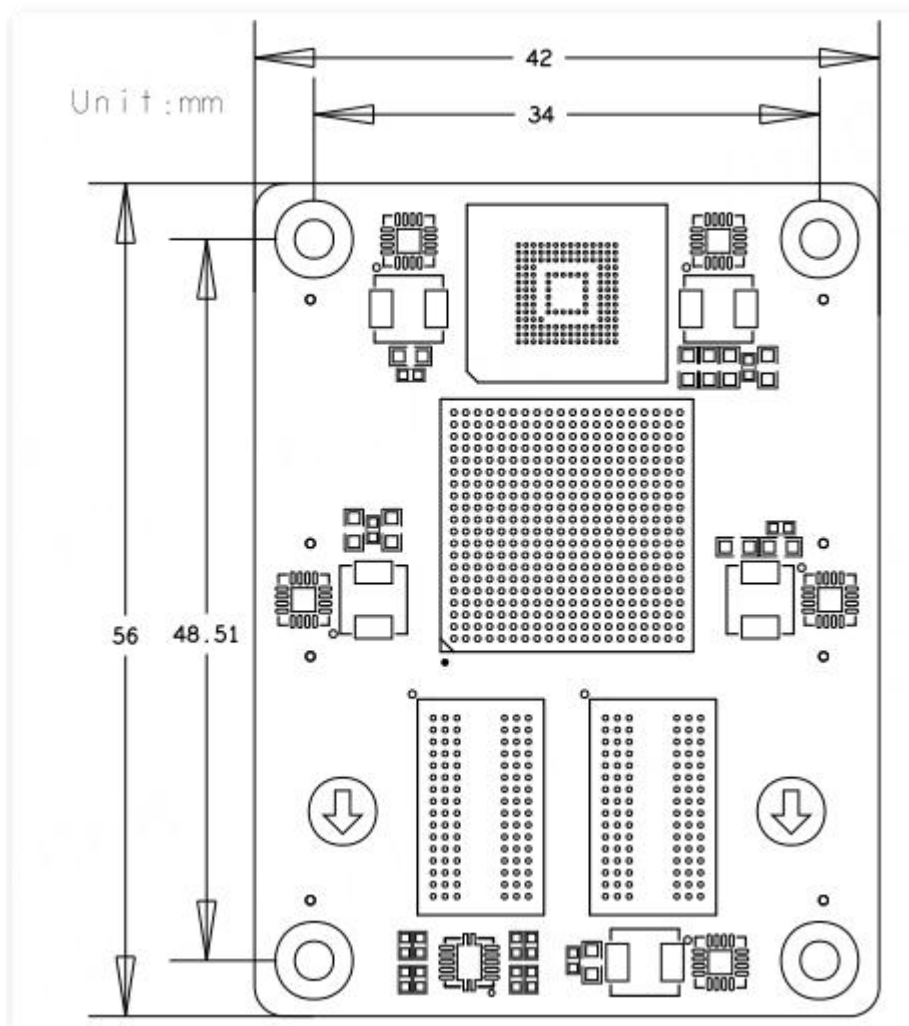


Figure 2.2-1 Haoyue_Development Board Core Board Front Mechanical Dimensions

2.3 Core board hardware resources

紫光 PG2L100H 核心板硬件资源	
尺寸	42*56 mm
主芯片	XC7Z010 封装: CLG400I XC7Z020 封装: CLG400I
PCB	8 层、黑色沉金
DDR3	DDR3*2, 型号: MT41K256M16, 挂载在 PS 端, 容量: 1GB, 位宽: 32bit
工作温度	-25℃~+85℃
Flash	型号: W25Q128, 容量: 32MB
晶振	PL 端 50MHZ 晶振, PS 端 33.3MHZ 晶振
EMMC	型号: KLM8G1GETF-B041, 容量: 8GB

2.3.1 DDR3

The core board carries two high-speed DDR3 SDRAM memory chips, model MT41K64M16TW-107, mounted on the PS side, with a single-chip capacity of 512MByte. In the design, we directly connected DDR3 to the memory interface of BANK 502 on the PS side.

The DDR3 hardware schematic diagram is shown in Figure 2.3-1 below:

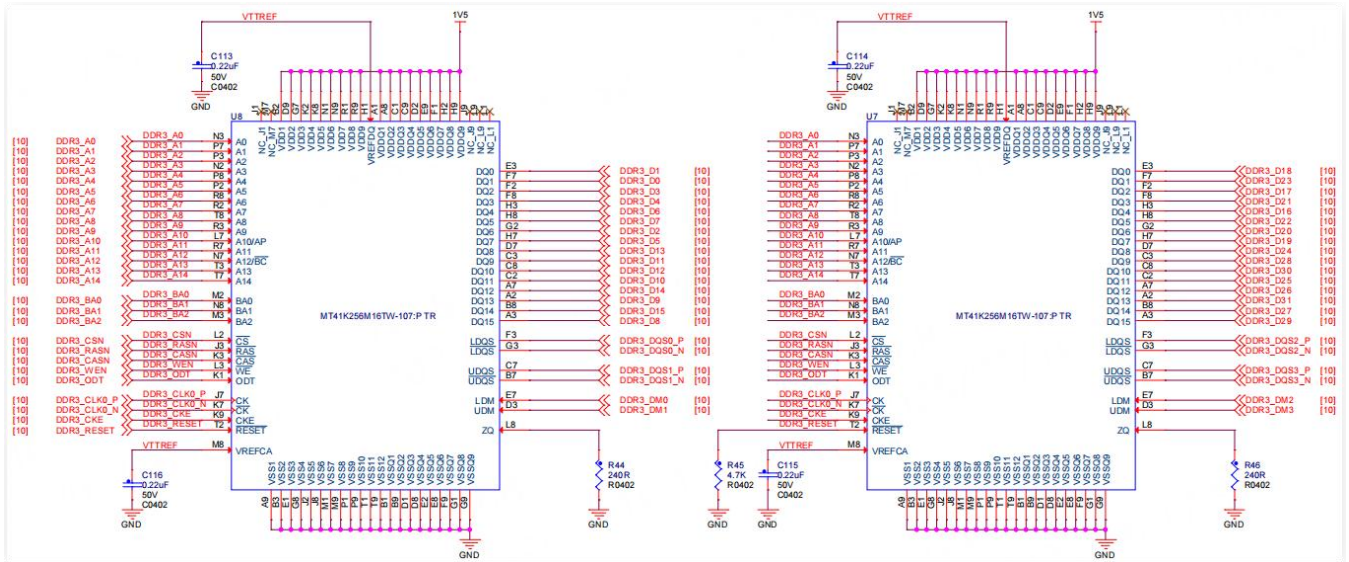


Figure 2.3-1 Partial schematic diagram of core board DDR3

The actual picture of DDR3 on the core board is shown in Figure 2.3-2 below:

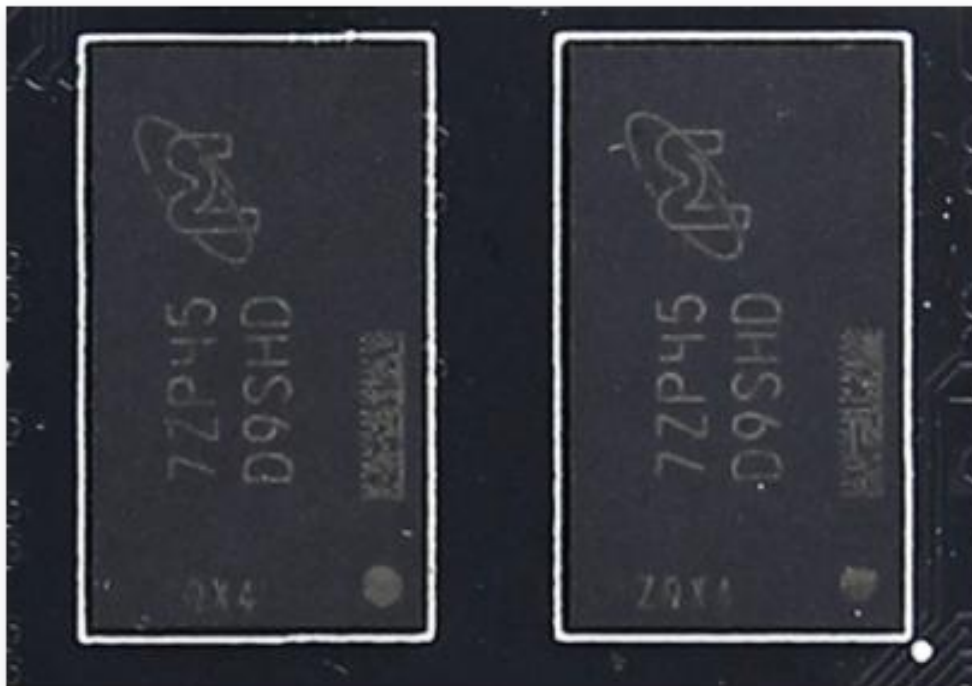


Figure 2.3-2 Actual picture of core board DDR3

Because DDR3 is mounted on the PS side, there is a BANK502 on the PS side dedicated to connecting DDR3. It has been designed when designing the schematic diagram. We only need to configure the DDR3 controller when doing related projects.

2.3.2 SPI-FLASH

The Haoyue core board is equipped with a FLASH memory chip, the chip model is W25Q128, and the single-chip storage capacity is 16MB. Since FLASH has the characteristic of not losing data when power is turned off, and our FPGA chip will lose data when it is powered off, we can use FLASH as the power-on configuration device of the FPGA chip. We solidify the power-on program in FLASH. After power-on, the FPGA chip reads the program stored in FLASH and runs it, so that the program is not lost when power is turned off. But unlike other FPGA development boards that only have the PL side, equipped with ZYNQ The flash of the series chip board is the pin connected to the PS side. The firmware needs to configure the PS side and connect the PL to the PS.

Figure 2.3-3 is the SPI_FLASH schematic diagram:

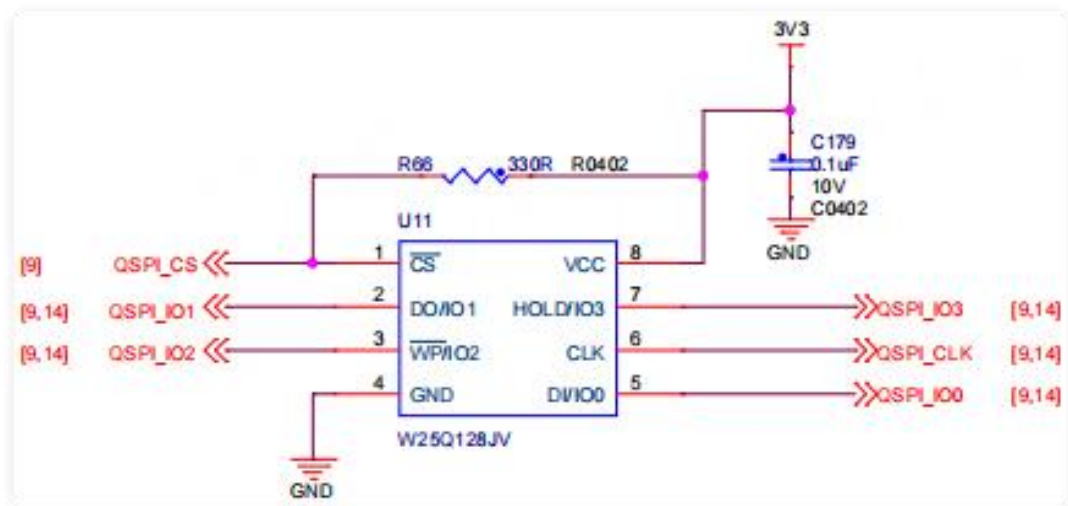


Figure 2.3-3 SPI_FLASH schematic diagram

Figure 2.3-4 is a physical picture of the FLASH chip on the Haoyue core board:

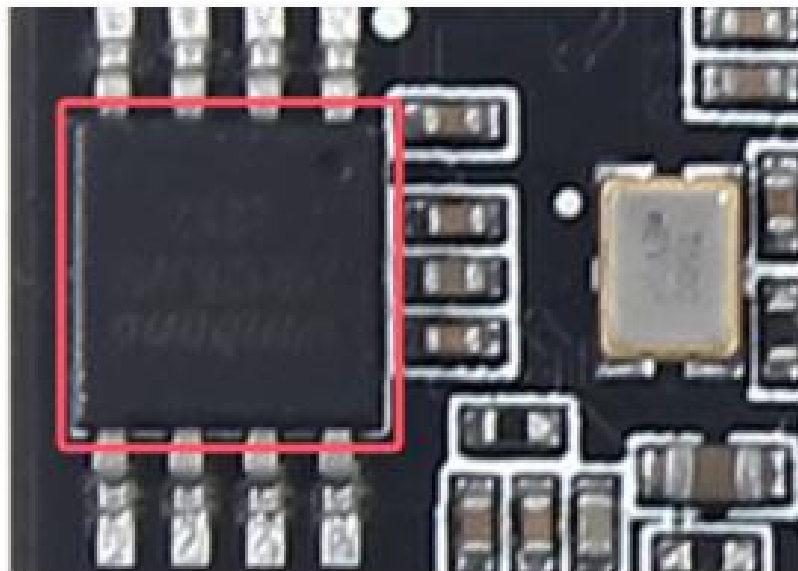


Figure 2.3-4 SPI_FLASH chip physical picture

2.3.3 Crystal oscillator

The Haoyue core board is equipped with a 50MHz active crystal oscillator. The output clock of the crystal oscillator is connected to the global clock pin of the FPGA. It is used as the system clock of the development board to drive the user logic circuit in the FPGA. At the same time, this clock can also be used to generate a higher or lower clock frequency for users to use through the clock IP core inside the FPGA.

Figure 2.3-5 is the circuit diagram of the Haoyue core board active crystal oscillator. You can see that an FPGA clock signal is output from this circuit diagram and connected to the FPGA chip for use by the FPGA.

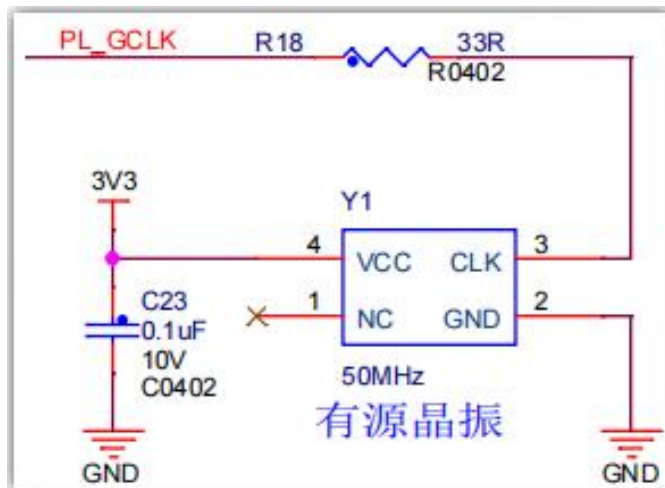


Figure 2.3-5 50MHz active crystal oscillator schematic diagram

Figure 2.3-6 is a physical picture of the 50MHz active crystal oscillator on the Haoyue core board.

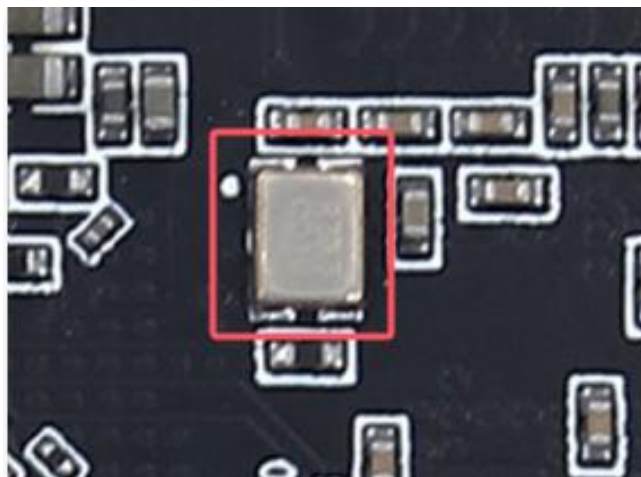


Figure 2.3.6 Actual picture of core board 50MHz active crystal oscillator

The 50MHz clock pin assignment table is shown in the following table:

引脚名	FPGA 绑定引脚
FPGA_CLK	N18

At the same time, the core board is also equipped with a 33.3MHz active crystal oscillator. This clock is used on the PS side

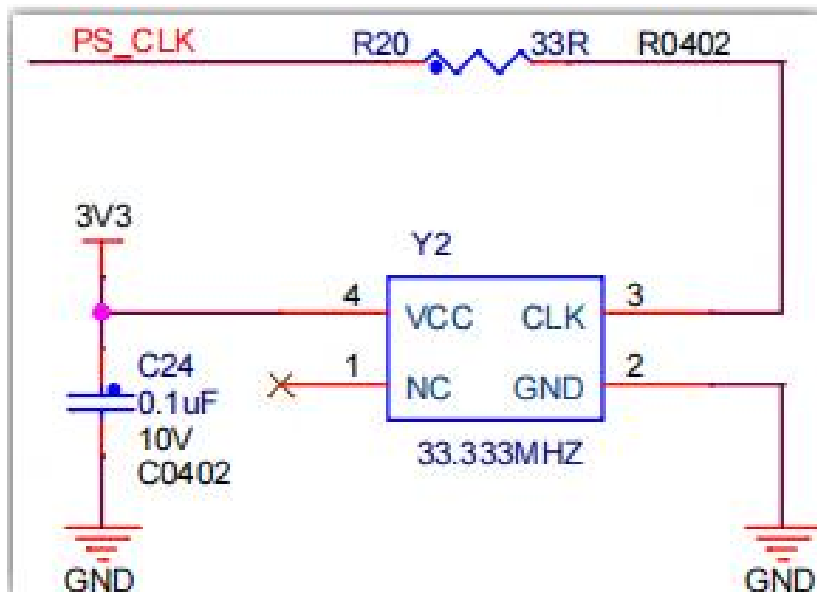


Figure 2.3-7 33.3MHz active crystal oscillator schematic diagram

Figure 2.3-8 is a physical picture of the 33.3MHz active crystal oscillator on the Haoyue core board.

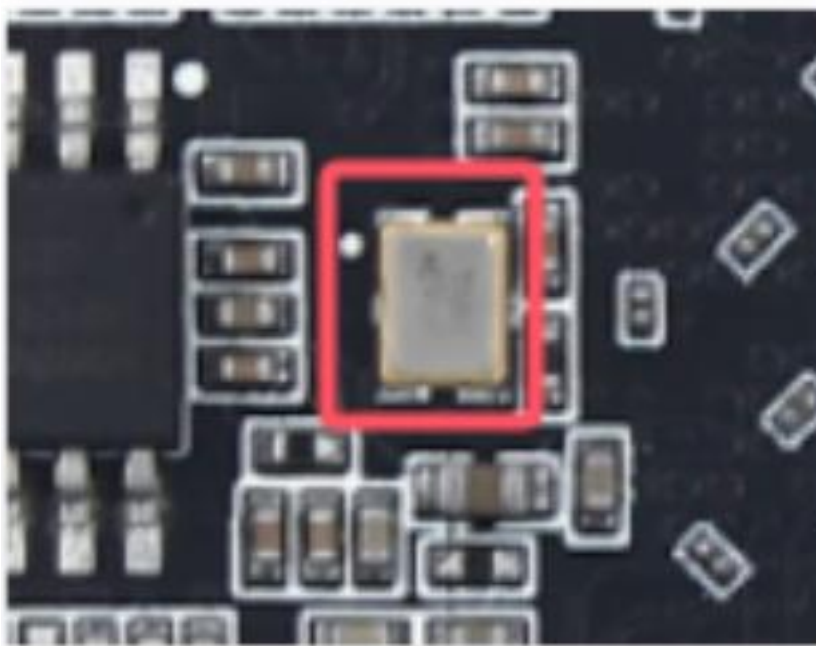


Figure 2.3-8 Actual picture of 33.3MHz active crystal

2.3.4 EMMC

The Haoyue core board is equipped with an eMMC chip, which is connected to the PS side of the core board. The model is KLM8G1GETF. It has a storage capacity of 8GByte and is suitable for storing user data. Figure 2.3-9 is the schematic diagram of the eMMC chip on the Haoyue core board.

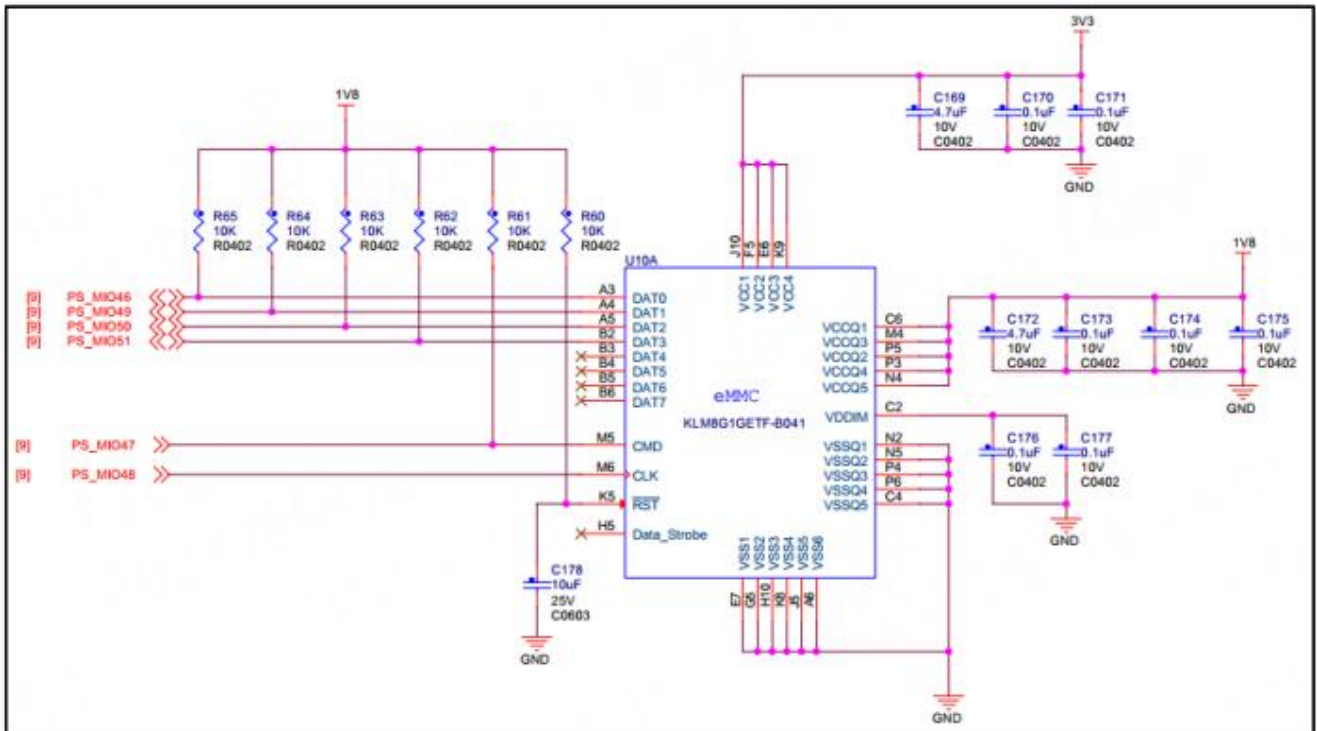


Figure 2.3-9 eMMC schematic diagram

Figure 2.3-10 is a physical picture of the eMMC on the Haoyue core board.

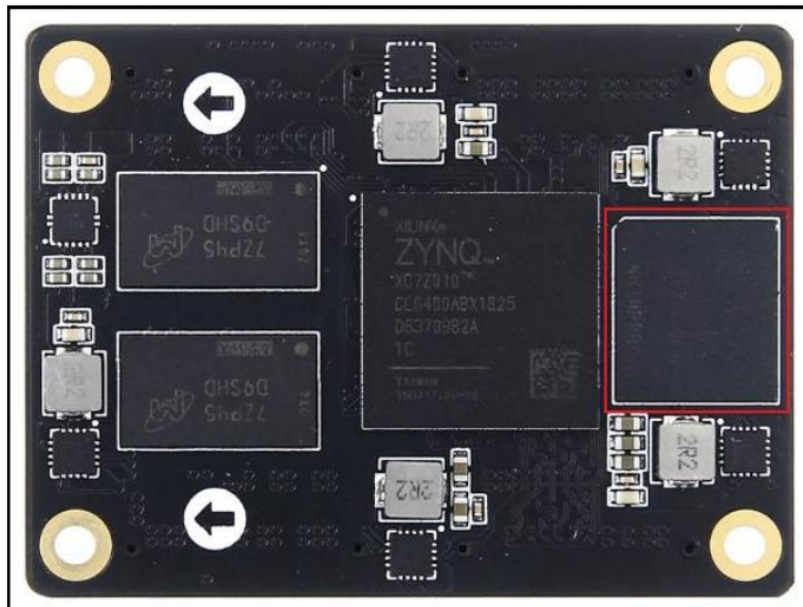


Figure 2.3-10 Physical picture of eMMC chip

2.3.4 Power supply

The power supply of Haoyue core board comes from the BTB base, and the input voltage is 12V. The core board power supply can be divided into two parts. The first part is converted into 1.0V, 1.8V, 1.5V, 3.3V, and VCCI O_35 through 5-channel DCDC chips. These five channels use TLV62130RGTR, and the current can reach 3A. Among them, BANK 35 uses a separate channel of VCCI O_35 for power supply. The VCCI O_35 voltage can be changed by changing the resistance of R14 and R15. The factory default is 3.3V. In addition, according to the official power supply design requirements, the power supply sequence of the first part has also been designed accordingly. The power-on sequence is 1.0V->1.8V->1.5V->3.3V->VCCI O_35. The second part is the VTT power supply and VTTREF reference power supply of DDR3, which uses the TPS51200DRCR chip, which can provide at least 3A sink current and source current for the DDR chip.

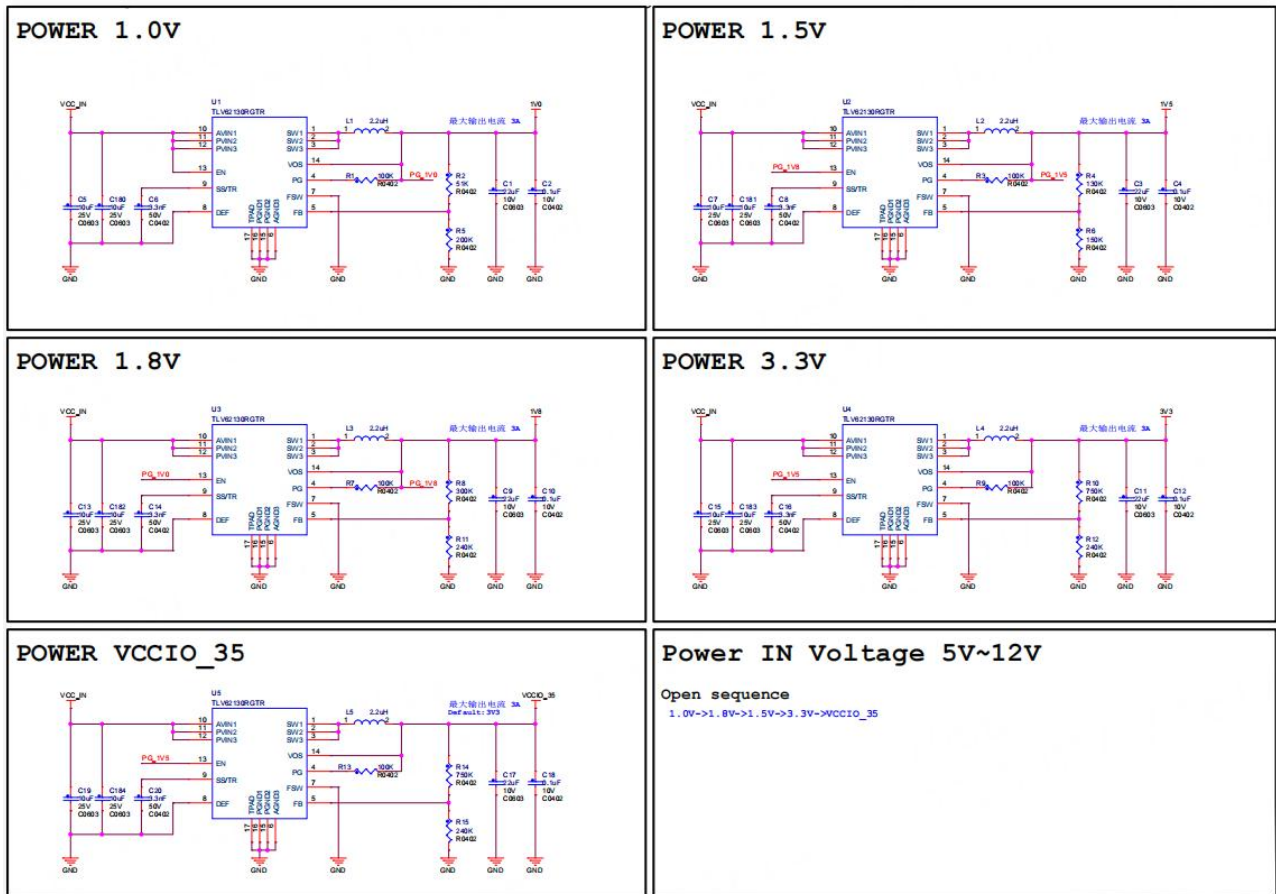


Figure 2.3-11 The first part of power supply

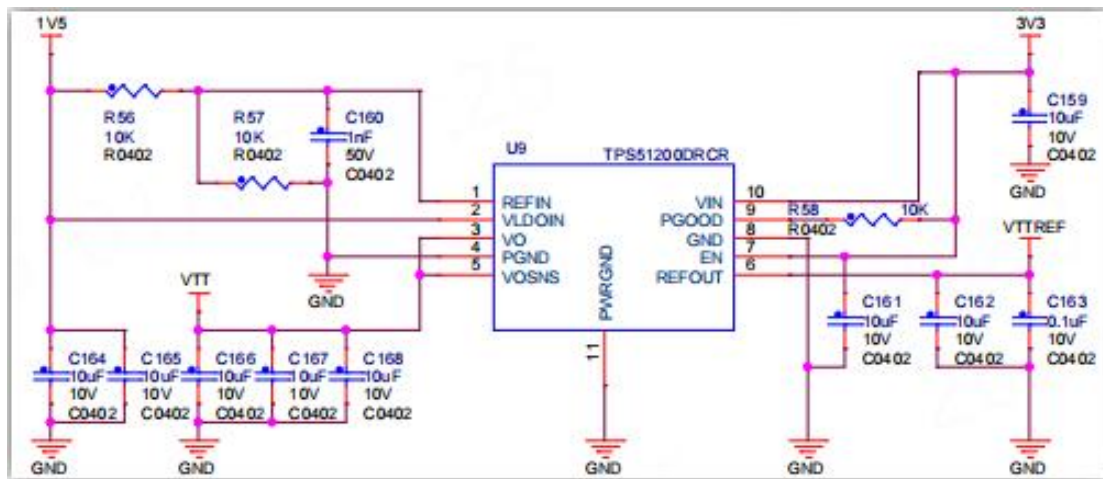


Figure 2.3-12 VTT power supply

引脚名	功能
1.0V	FPGA 核心电源
1.8V	FPGA 辅助电源
1.5V	FPGA 的 BANK 502 和 DDR3
3.3V	FPGA 的 BANK0、13、34、35 以及 FLASH、有源晶振
VCCIO_35	FPGA 的 BANK 35
VTT、VTTREF(0.75)	DDR3

2.3.5 BTB connector

The core board has four 60-pin BTB interfaces onboard to lead out the IO ports with an interval of 0.5mm. The core board is connected to the baseboard through this interface. The physical diagram of the interface is shown in Figure 2.3-13.

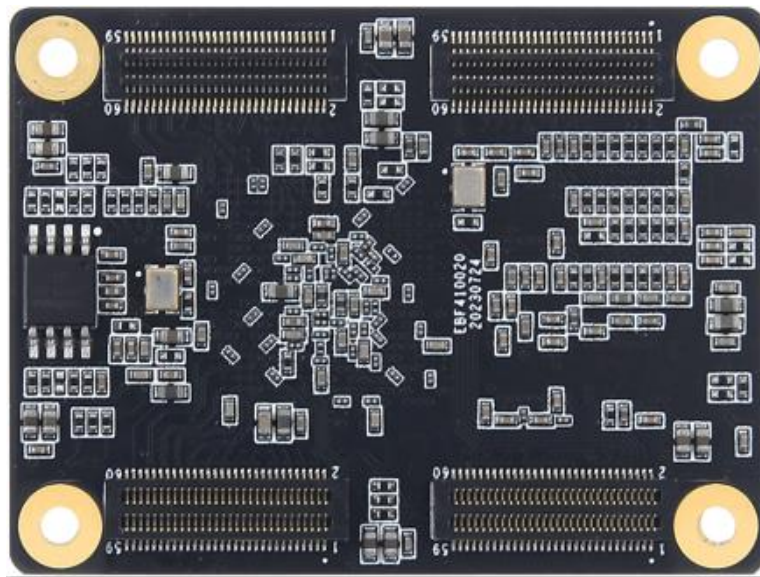


Figure 2.3-13 BTB interface physical diagram

The CN1 pin assignment is shown in the table below:

编号	引脚名	FPGA 绑定引脚	编号	引脚名	FPGA 绑定引脚
1	GND	GND	2	GND	12V
3	IO34_L4P	V12	4	IO34_L1P	T11
5	IO34_L4N	W13	6	IO34_L1N	T10
7	IO34_L10P	V15	8	IO34_L2P	T12
9	IO34_L10N	W15	10	IO34_L2N	U12
11	IO34_0	R19	12	IO34_L3P	U13
13	IO34_L13N	P19	14	IO34_L3N	V13
15	GND	GND	16	IO34_L5P	T14
17	IO34_L8P	W14	18	IO34_L5N	T15
19	IO34_L8N	Y14	20	GND	GND
21	IO34_L7P	Y16	22	IO34_L21P	V17
23	IO34_L7N	Y17	24	IO34_L21N	V18

25	IO34_L17P	Y18	26	IO34_L9P	T16
27	IO34_L17N	Y19	28	IO34_L9N	U17
29	IO34_L16P	V20	30	IO34_L24P	P15
31	IO34_L16N	W20	32	IO34_L24N	P16
33	IO34_L15P	T20	34	IO34_L6P	P14
35	IO34_L15N	U20	36	IO34_L6N	R14
37	GND	GND	38	GND	GND
39	IO34_L11P	U14	40	IO34_L18P	V16
41	IO34_L11N	U15	42	IO34_L18N	W16
43	GND	GND	44	IO34_L19P	R16
45	IO35_L10P	K19	46	IO34_L19N	R17
47	IO35_L10N	J19	48	IO34_L22P	W18
49	GND	GND	50	IO34_L22N	W19
51	IO35_L13P	H16	52	IO34_L12P	U18
53	IO35_L13N	H17	54	IO34_L12N	U19
55	IO35_L14P	J18	56	GND	GND
57	IO35_L14N	H18	58	IO34_25	T19
59	GND	GND	60	GND	GND

The CN2 pin assignment is shown in the table below:

编号	引脚名	FPGA 绑定引脚	编号	引脚名	FPGA 绑定引脚
1	GND	12V	2	GND	GND
3	IO35_L23N	12V	4	PS_MIO52	C10
5	IO35_L23P	12V	6	PS_MIO53	C11
7	IO35_L22N	12V	8	IO35_IO0	G14
9	IO35_L22P	N12	10	IO35_IO25	J15
11	IO35_L21N	GND	12	GND	GND
13	IO35_L21P	AA9	14	IO35_L24N	J16
15	GND	AB10	16	IO35_L24P	K16
17	IO35_L18N	AA10	18	GND	GND
19	IO35_L18P	AA11	20	IO35_L20N	J14
21	IO35_L17N	GND	22	IO35_L20P	K14
23	IO35_L17P	Y11	24	GND	GND
25	GND	Y12	26	IO35_L19N	G15
27	IO35_L16N	V13	28	IO35_L19P	H15

29	IO35_L16P	V14	30	GND	GND
31	IO35_L15N	GND	32	IO35_L6N	F17
33	IO35_L15P	AA13	34	IO35_L6P	F16
35	GND	AB13	36	IO35_L5N	E19
37	IO35_L12N	W14	38	IO35_L5P	E18
39	IO35_L12P	Y14	40	GND	GND
41	IO35_L11N	GND	42	IO35_L4N	D20
43	IO35_L11P	W15	44	IO35_L4P	D19
45	GND	W16	46	IO35_L3N	D18
47	IO35_L9N	T16	48	IO35_L3P	E17
49	IO35_L9P	U16	50	GND	GND
51	IO35_L8N	GND	52	IO35_L2N	A20
53	IO35_L8P	AB16	54	IO35_L2P	B19
55	IO35_L7N	AB17	56	IO35_L1N	B20
57	IO35_L7P	GND	58	IO35_L1P	C20
59	GND	GND	60	GND	GND

The CN3 pin assignment is shown in the table below:

编号	引脚名	FPGA 绑定引脚	编号	引脚名	FPGA 绑定引脚
1	GND	GND	2	GND	GND
3	IO34_L14N	P20	4	FPGA_TCK	F9
5	IO34_L14P	N20	6	FPGA_TDO	F6
7	IO34_L20N	R18	8	FPGA_TDI	G6
9	IO34_L20P	T17	10	FPGA_TMS	J6
11	IO34_L23N	P18	12	GND	GND
13	IO34_L23P	N17	14	IO13_L6N	V5
15	GND	GND	16	GND	GND
17	BOOT_MIO5	A6	18	PS_POR_B	C7
19	BOOT_MIO4	B7	20	GND	GND
21	GND	GND	22	FPGA_DONE	R11
23	IO13_L22N	W6	24	GND	GND
25	IO13_L22P	V6	26	XADC_VN	L10
27	IO13_L21N	V10	28	XADC_VP	K9
29	IO13_L21P	V11	30	GND	GND
31	GND	GND	32	IO13_L15N	W8

33	IO13_L20N	Y13	34	IO13_L15P	V8
35	IO13_L20P	Y12	36	GND	GND
37	GND	GND	38	IO13_L14N	Y8
39	IO13_L19N	U5	40	IO13_L14P	Y9
41	IO13_L19P	T5	42	GND	GND
43	GND	GND	44	IO13_L13N	Y6
45	IO13_L18N	Y11	46	IO13_L13P	Y7
47	IO13_L18P	W11	48	GND	GND
49	GND	GND	50	IO13_L12N	U10
51	IO13_L17N	U8	52	IO13_L12P	T9
53	IO13_L17P	U9	54	GND	GND
55	IO13_L16N	W9	56	IO13_L11N	V7
57	IO13_L16P	W10	58	IO13_L11P	U7
59	GND	GND	60	GND	GND

The CN4 pin assignment is shown in the table below:

编号	引脚名	FPGA 绑定引脚	编号	引脚名	FPGA 绑定引脚
1	VCC_IN	12V	2	VCC_IN	12V
3	VCC_IN	12V	4	VCC_IN	12V
5	VCC_IN	12V	6	VCC_IN	12V
7	VCC_IN	12V	8	VCC_IN	12V
9	GND	GND	10	GND	GND
11	GND	GND	12	GND	GND
13	GND	GND	14	GND	GND
15	GND	GND	16	GND	GND
17	PS_MIO0	E6	18	PS_MIO40	D14
19	PS_MIO7	D8	20	PS_MIO41	C17
21	PS_MIO8	D5	22	PS_MIO42	E12
23	PS_MIO9	B5	24	PS_MIO43	A9
25	PS_MIO10	E9	26	PS_MIO44	F13
27	PS_MIO11	C6	28	PS_MIO45	B15
29	PS_MIO14	C5	30	PS_MIO12	D9
31	PS_MIO15	C8	32	GND	GND
33	PS_MIO16	A19	34	PS_MIO28	C16

35	PS_MIO17	E14	36	PS_MIO29	C13
37	PS_MIO18	B18	38	PS_MIO30	C15
39	PS_MIO19	D10	40	PS_MIO31	E16
41	PS_MIO20	A17	42	PS_MIO32	A14
43	PS_MIO21	F14	44	PS_MIO33	D15
45	GND	GND	46	PS_MIO34	A12
47	PS_MIO22	B17	48	PS_MIO35	F12
49	PS_MIO23	D11	50	PS_MIO36	A11
51	PS_MIO24	A16	52	PS_MIO37	A10
53	PS_MIO25	F15	54	PS_MIO38	E13
55	PS_MIO26	A15	56	PS_MIO39	C18
57	PS_MIO27	D13	58	PS_MIO13	E8
59	GND	GND	60	GND	GND

Chapter 3 Introduction to Haoyue Development Board B

3.1 Appearance of base plate

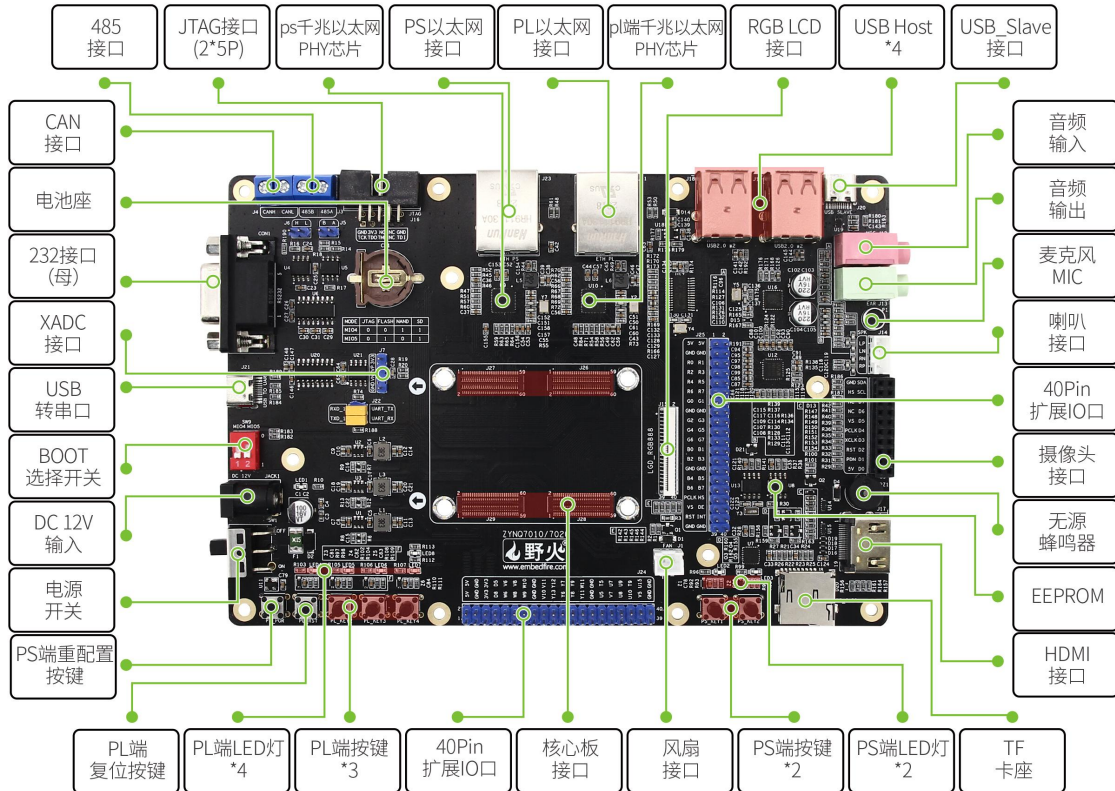


Figure 3.1-1 Front view of Haoyue development board base plate

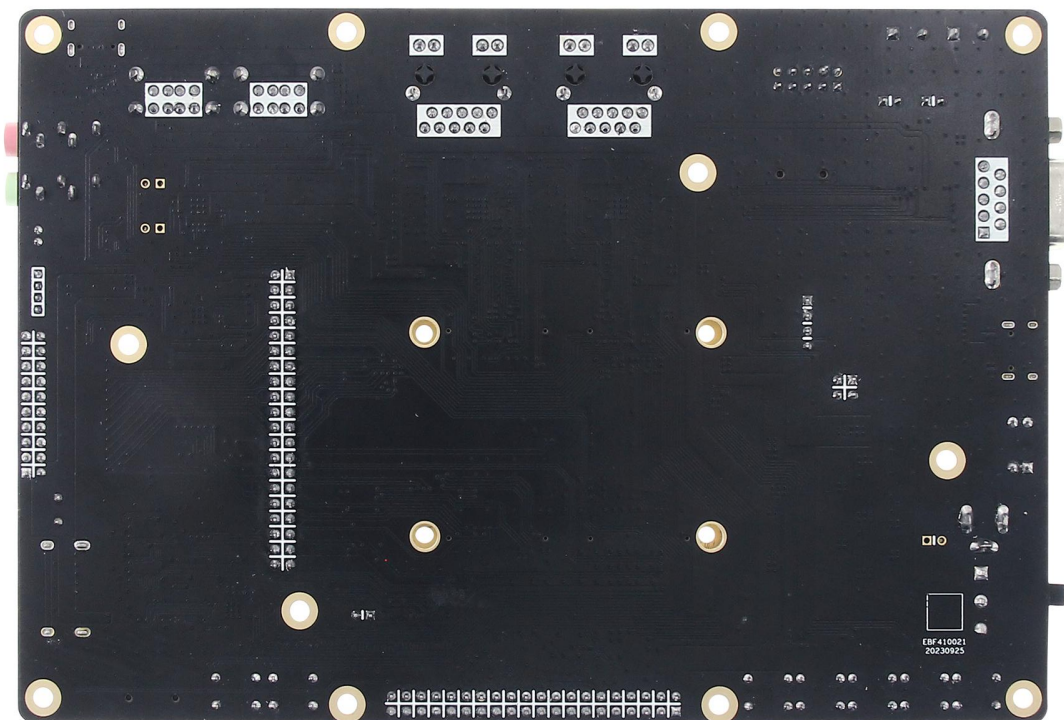


Figure 3.1-2 Back view of Haoyue development board base plate

3.2 Bottom plate dimension drawing

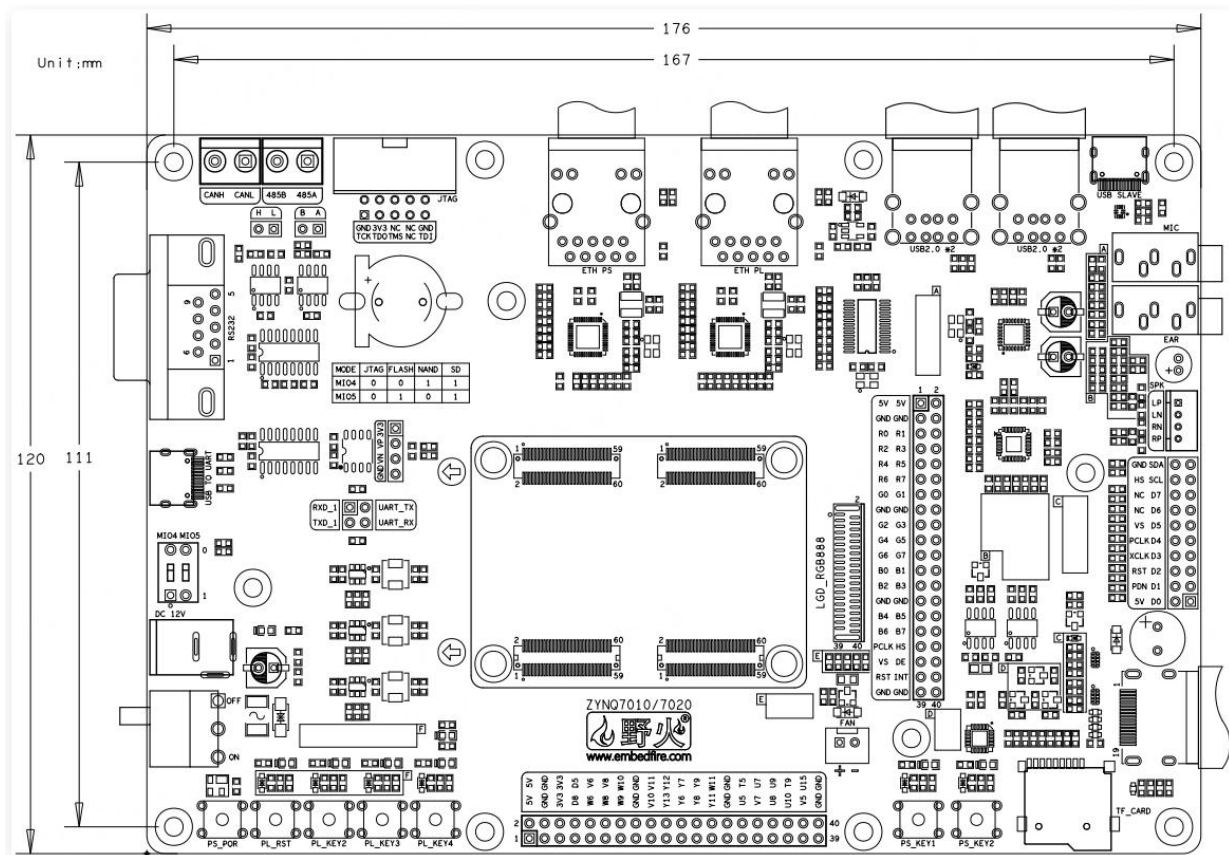


Figure 3.2-1 Mechanical dimensions of the front side of the Haoyue development board base

3.3 Backplane hardware specifications

凌云开发板底板硬件规格	
尺寸	176*120MM
PCB	4 层、黑色沉金
电源输入	12V@1A 直流输入，DC 接口
KEY	PL 端复位按键*1，PS 端重配置按键*1， PL 端功能按键*3，PS 端功能按键*2
单色用户灯	PL 端 LED*4，PS 端 LED*2
EEPROM	型号：M24C64，容量：64Kbit
USB 转串口	1 个，使用Type-C 接口
以太网	PS 端千兆以太网*1，PL 端千兆以太网*1， PHY 芯片型号：RTL8211E
JTAG 接口	1 个，使用2*5P 2.54 间距牛角插座出
摄像头接口	1 个，可接野火单目 OV5640 摄像头/OV7725 摄像头（不带 FIFO）

蜂鸣器	无源蜂鸣器*1, 3.3V 驱动
HDMI	HDMI Type-A 接口*1
LCD 接口	40PinFPC 接口*1, 0.5mm 间距 RGB 接口时序, 支持 RGB565/888 屏幕
SD	底板 Micro SD (TF) 卡座*1
RTC	RTC 电池座*1, 实时时钟芯片: PCF8563T
喇叭接口	1 个, 1*4Pin 接口引出
USB2.0	USB HOST 接口*4, USB SLAVE 接口*1
音频	音频输入, 3.5mm 音频座*1 (红); 音频输出, 3.5mm 音频座*1 (绿); 音频解码芯片: WM8960; 电容咪头*1
开关	BOOT 模式选择开关*1
232	1 路母头 D89
485	1 路, 2 线接线柱引出, 5mm 间距
CAN	1 路, 2 线接线柱引出, 5mm 间距
XADC	XADC 接口*1

3.4 Instructions for using baseboard hardware

We will introduce in detail the hardware resources of each part of the Haoyue development board baseboard (Figure 3.1-1 Lingyun Pro baseboard hardware resource diagram), and we will introduce them in turn.

3.4.1 Power supply

The main power supply method of the Haoyue development board baseboard is DC 12V power input. It is recommended to use a 12V power supply for the development board. Try to avoid using power supplies of other specifications during use to avoid damaging the development board.

Figure 3.4-1 is the power input circuit design of the Haoyue development board backplane:

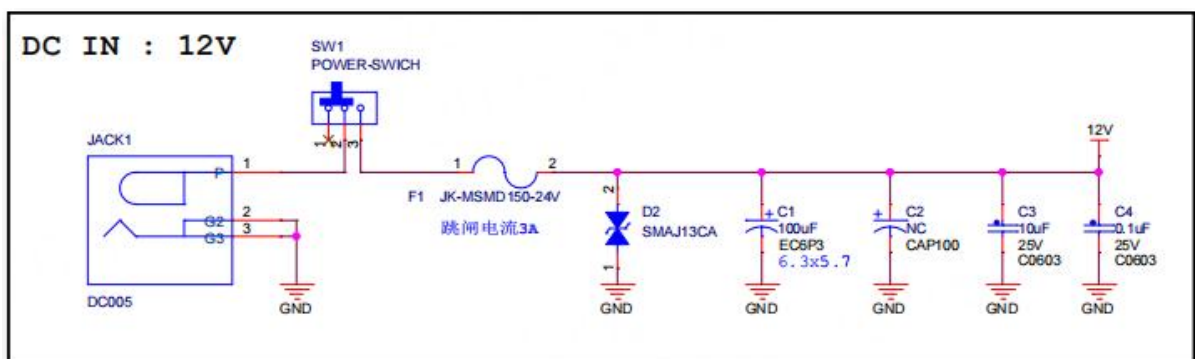


Figure 3.4-1 Power input

The baseboard converts 5V, 3.3V, and 1.8V through the three-channel DCDC chip TP6327GS6, and the current can reach 3A, providing sufficient power to the baseboard peripherals. The schematic diagram is shown in Figure 3.3-2 below:

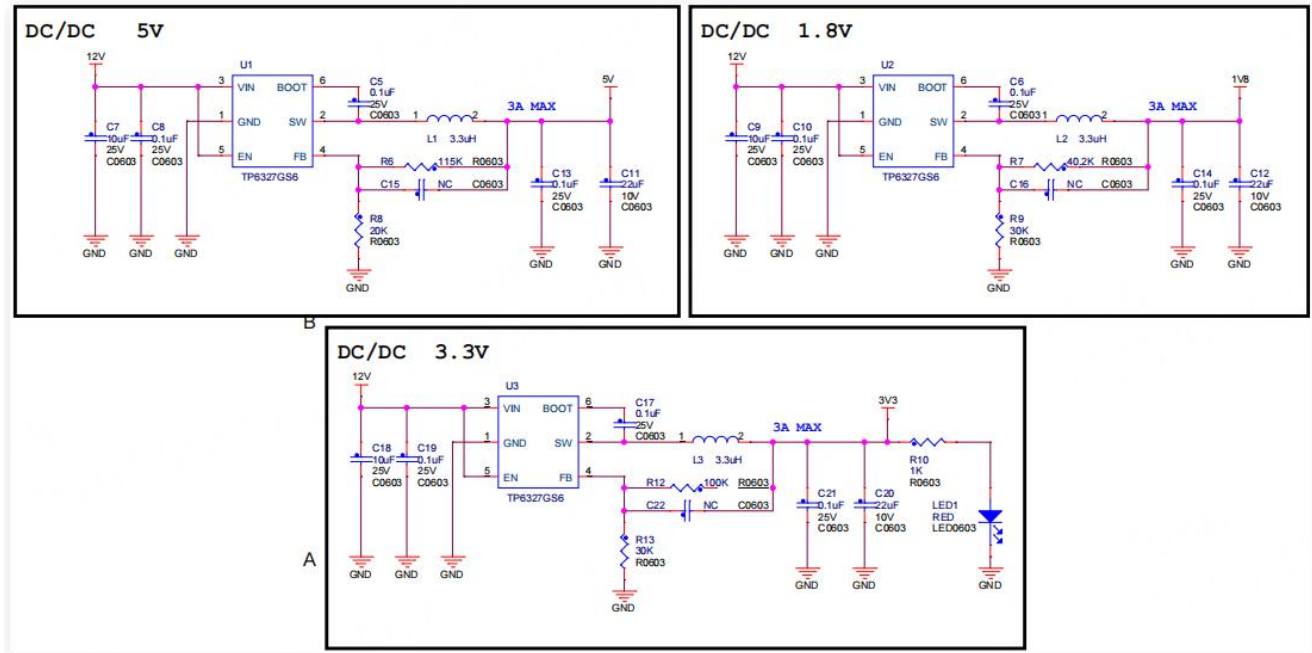


Figure 3.4-2 Three-way DCDC power supply

The distribution functions of each power supply on the Haoyue development board backplane are as shown in the following

电源	功能
5V	CAN、LCD、摄像头接口等
3.3V	SD卡、EEPROM、HDMI、以太网等
1.8V	SD卡、以太网等

3.4.2 EEPROM

The Haoyue development board baseboard carries an EEPROM chip, model M24C64, with a capacity of 64K bit (8Kbyte). FPGA communicates with it through the IIC bus, which is readable and writable. It has the characteristics of not losing data when power is turned off. It can be used to store some important data that cannot be lost when power is turned off, such as chip configuration data.

Figure 3.3-3 is the schematic diagram of the EEPROM chip on the backplane of Haoyue development board:

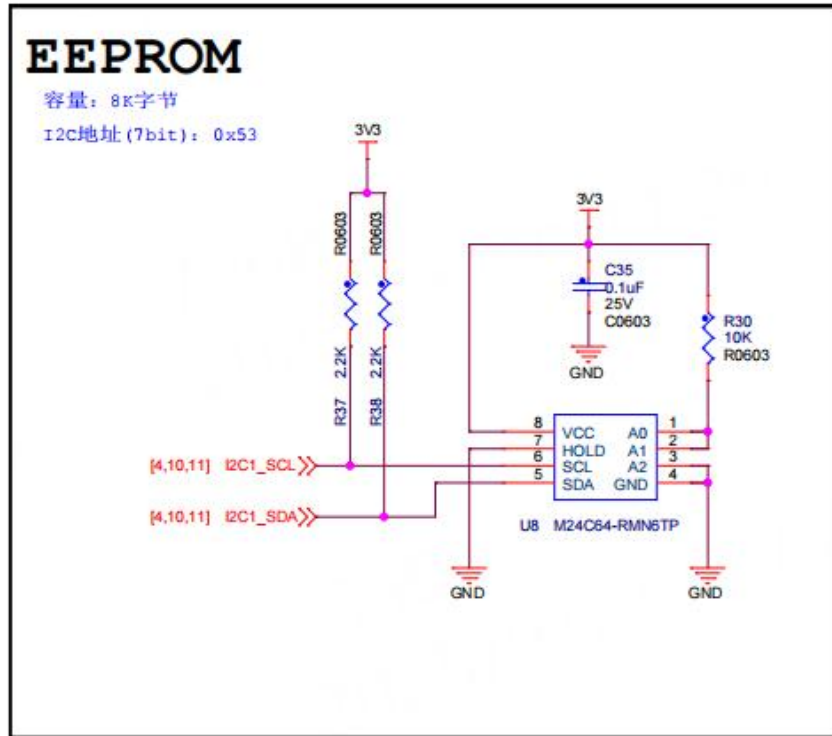


Figure 3.4-3 Development board EEPROM schematic diagram

The physical picture of the EEPROM chip on the backplane of the development board is shown in Figure 3.4-4.

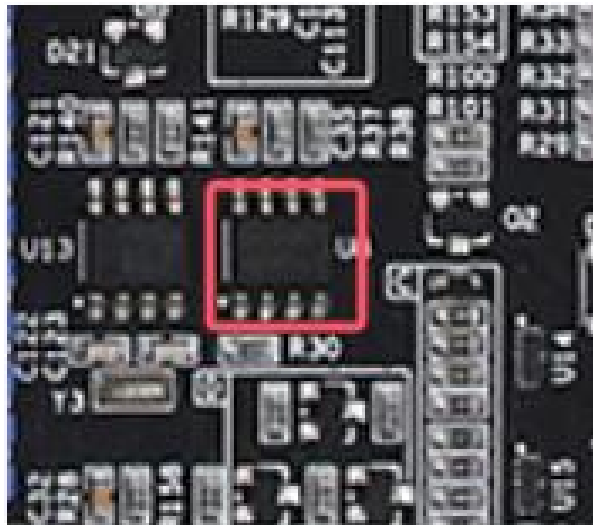


Figure 3.4-4 Physical picture of development board EEPROM

The pin assignments of EEPROM are as shown in the following table:

引脚名	FPGA 绑定引脚
I2C1_SCL	N17
I2C1_SDA	P18

3.4.3 HDMI input and output interface

The base board of the Haoyue development board is equipped with an HDMI input and output interface, which means that this port can be used for input and output, and supports up to 1080P@60Hz output display.

The HDMI interface schematic diagram is shown in Figure 3.4-5:

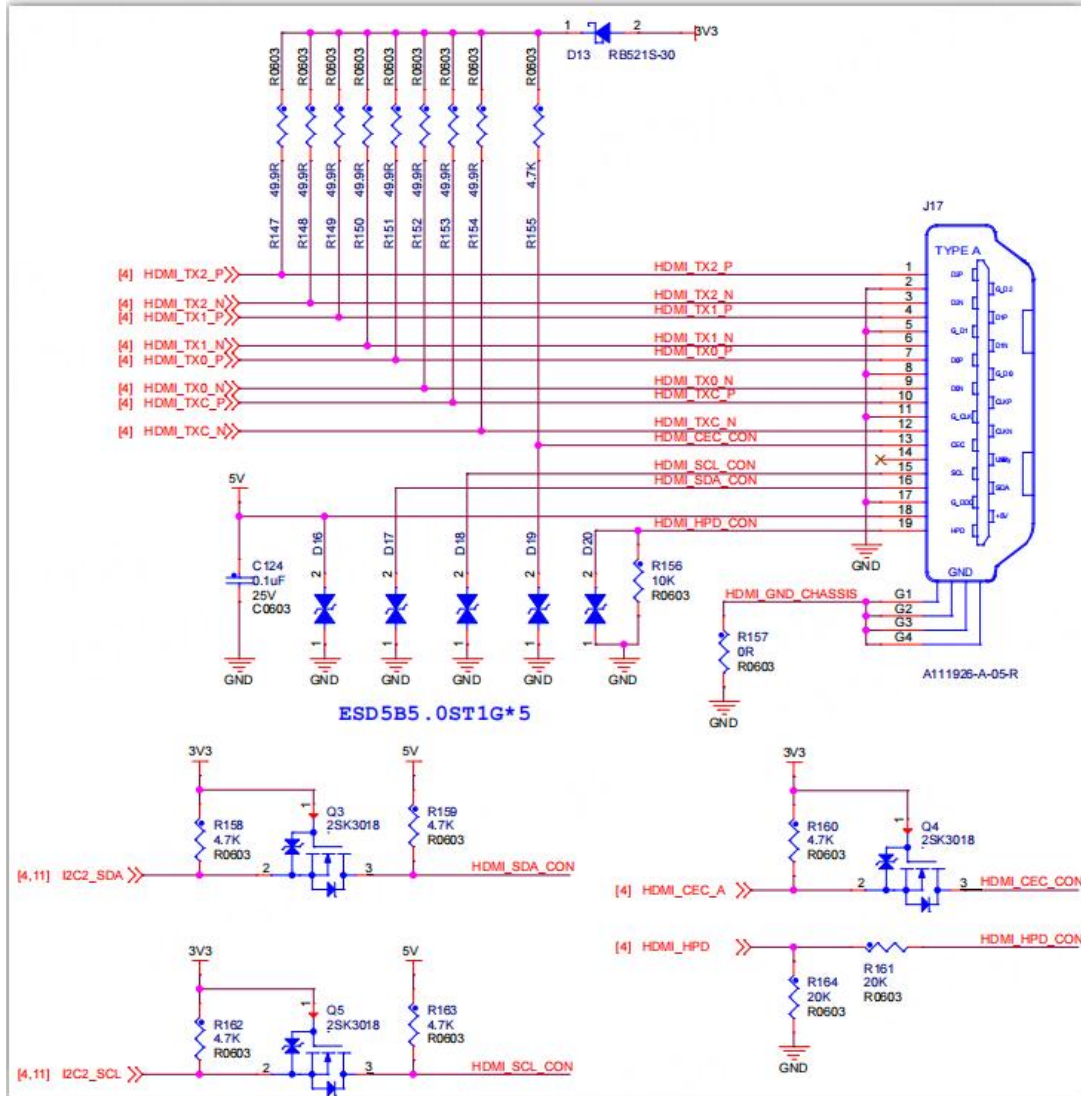


Figure 3.4-5 HDMI schematic diagram

The HDMI interface on the backplane of the Haoyue development board is shown in Figure 3.4-6, and the interface type is TYPE A.



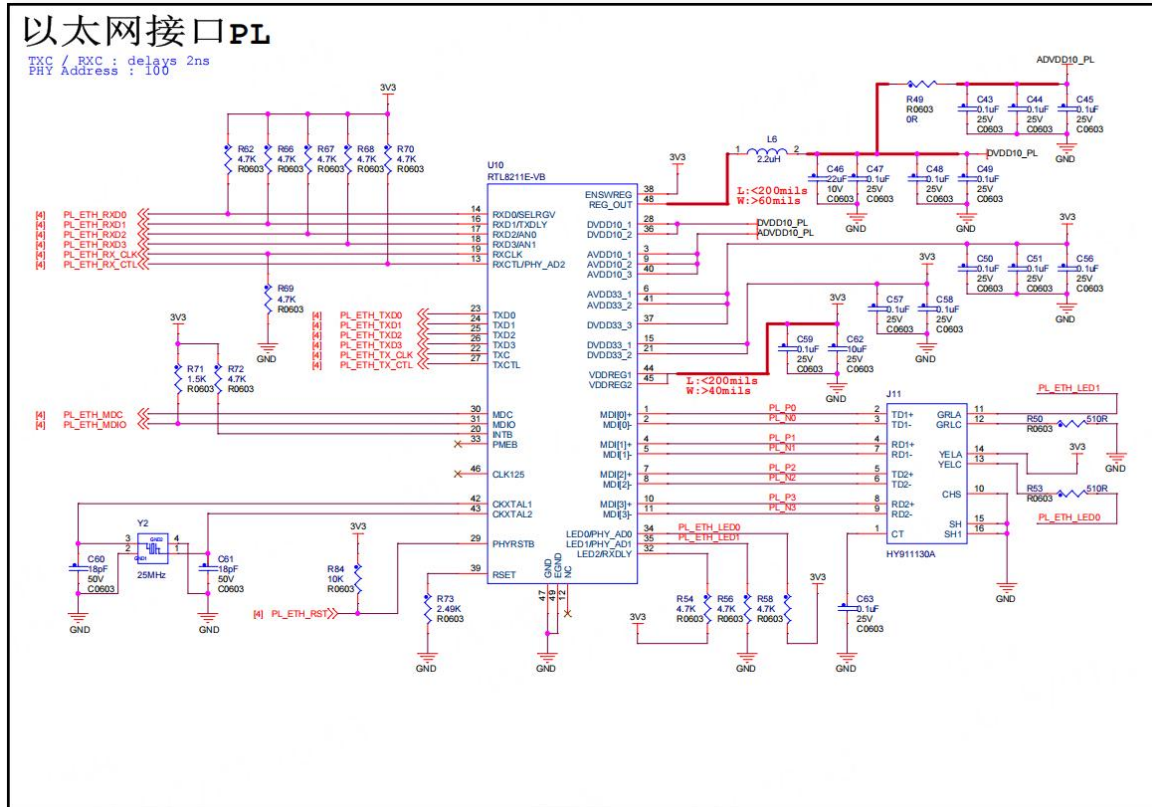
Figure 3.4-6 Actual picture of HDMI interface

The HDMI interface pin assignment is shown in the following table:

引脚名	FPGA 绑定引脚
HDMI_TX2_P	Y18
HDMI_TX2_N	Y19
HDMI_TX1_P	Y16
HDMI_TX1_N	Y17
HDMI_TX0_P	W14
HDMI_TX0_N	Y14
HDMI_TXC_P	V20
HDMI_TXC_N	W20
HDMI_CEC_A	T20
I2C2_SCL	P20
I2C2_SDA	N20
HDMI_HPD	U20

3.4.4 Gigabit Ethernet interface

The Haoyue development board has two Gigabit Ethernet interfaces on the backplane, one of which is mounted on the PS side and each communicates via the RTL8211E Ethernet PHY chip for Ethernet communication. RTL8211F is a low-power 10-BASE/100-BASE/1000-BASE full-duplex Ethernet PHY chip that supports 10Mbps, 100Mbps and 1000Mbps. The IO pin voltage is variable, compliant with the IEEE802.3-2005 standard, and supports Ethernet MAC through the RGMII interface. It can achieve the best connection method with the destination host through self-negotiation (adjust the best speed and duplex mode), and has HP Auto-MDIX automatic flip function to achieve direct connection or cross connection. The schematic diagram of RTL8211F is as follows:



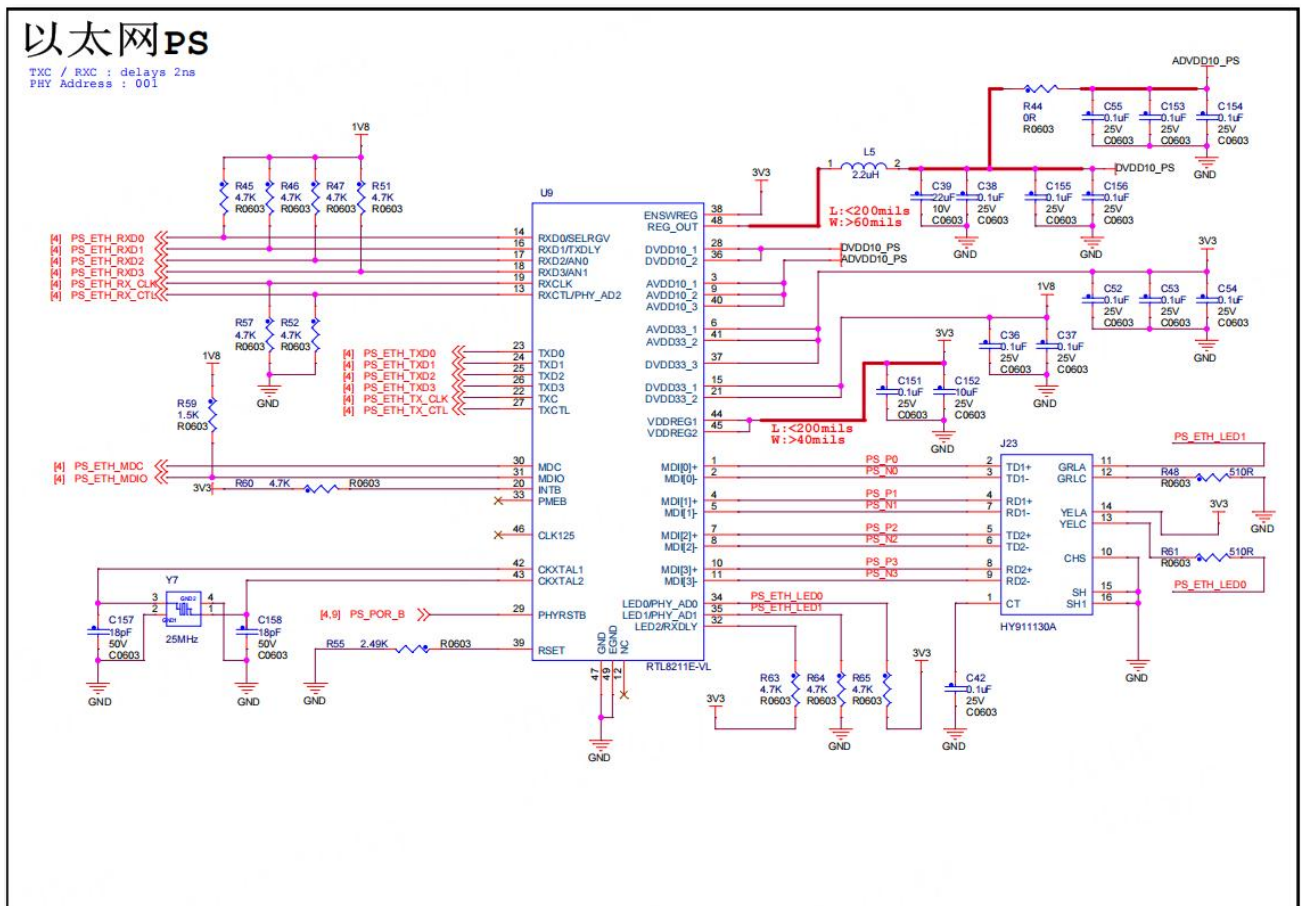


Figure 3.4-8 Partial schematic diagram of Ethernet PS

The physical diagram of Ethernet in the development board is shown in Figure 3.4-9:

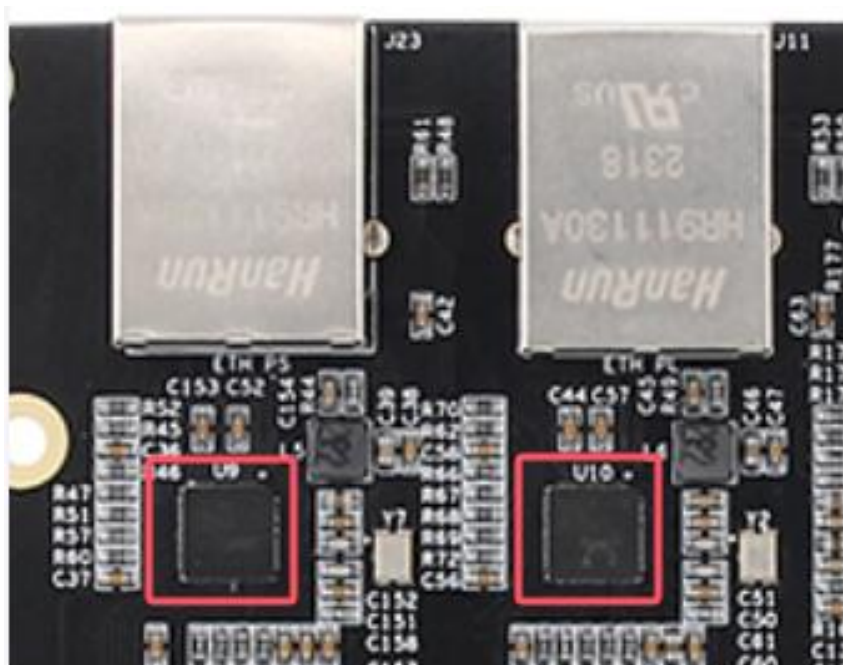


Figure 3.4-9 Physical diagram of Ethernet interface

The PL side Ethernet pin assignment is shown in the following table:

引脚名	FPGA 绑定引脚
PL_ETH_RXD0	E17
PL_ETH_RXD1	D18
PL_ETH_RXD2	F19
PL_ETH_RXD3	F20
PL_ETH_RX_CLK	K17
PL_ETH_RX_CTL	F17
PL_ETH_TXD0	E18
PL_ETH_TXD1	E19
PL_ETH_TXD2	G17
PL_ETH_TXD3	G18
PL_ETH_TX_CTL	F16
PL_ETH_TX_CLK	K18
PL_ETH_MDC	J20
PL_ETH_MDIO	H20
PL_ETH_INIB	AD26
PL_ETH_RST	J15

3.4.5 SD card interface

The Haoyue development board has an onboard SD card interface located on the PS side of the development board. It is used to access Micro SD cards and can read and write SD card data. SD was jointly launched by Panasonic, Toshiba and SanDisk and released in August 1998. Due to its small size, fast data transmission speed, hot-swappable and other excellent features, it is widely used in portable devices, such as digital cameras, smart phones, etc.

The schematic diagram of the SD card interface is shown in Figure 3.4-10:

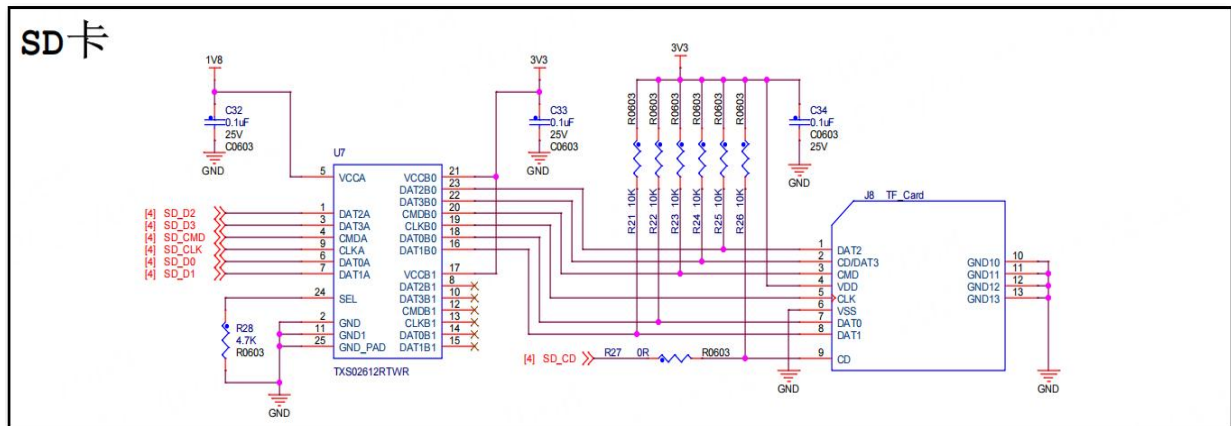


Figure 3.4-10 SD card interface schematic diagram

The physical picture of the SD card slot on the development board is shown in Figure 3.4-11:

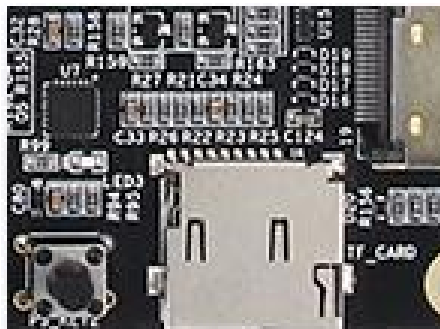


Figure 3.4-11 Physical diagram of SD card interface

3.4.6 JTAG interface

How do we download the software program to the FPGA chip? The JTAG interface on the development board is this bridge. The function of JTAG is to download the program file (.bit file) compiled by the software to the FPGA chip. However, after the file is downloaded to the FPGA, if the power is turned off and then on, the program will be lost and must be downloaded again. In the previous introduction to the core board, we explained FLASH. FLASH has the feature of not losing data when power is turned off, so we can configure the program file into a FLASH configuration program, and then solidify the file into FLASH. When the FPGA is powered on, the FPGA will read the program in FLASH and run it, so as to achieve the purpose of not losing data when power is turned off.

As shown in Figure 3.4-12, it is the schematic diagram of the JTAG interface:

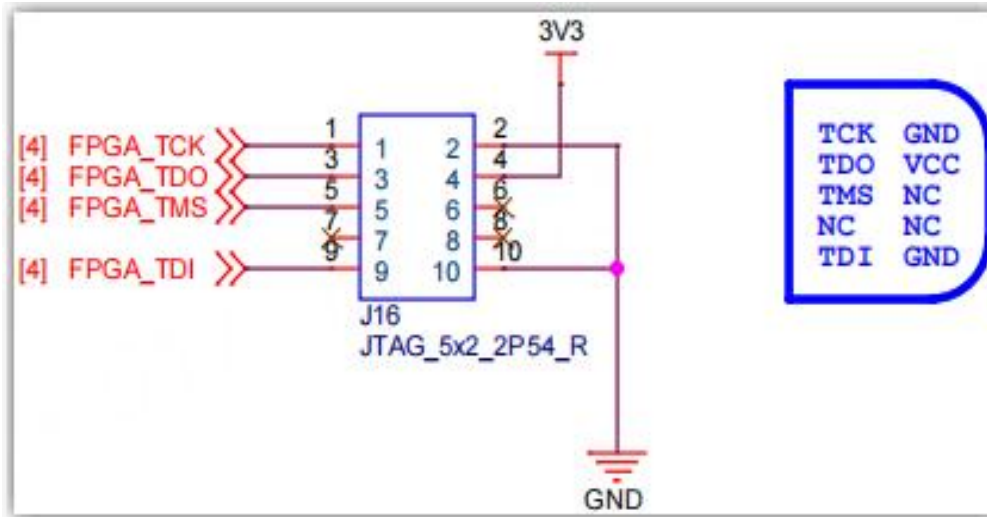


Figure 3.4-12 JTAG interface schematic diagram

Figure 3.4-13 Physical picture of the JTAG port on the Haoyue development board base plate:



Figure 3.4-13 Physical diagram of JTAG interface

Note: Try not to plug or unplug the JTAG port while the power is on, otherwise it is easy to burn out the JTAG port of the FPGA. If you use a multimeter to detect that any of the JTAG numbers TDI TDO TMS TCK is short-circuited to ground, your FPGA may have been burned out. Not every time the JTAG port is hot-swapped, it will definitely burn out, but at least there is a possibility of burning it out. In order to allow the development board to accompany us through this tutorial, it is best to proceed with caution!

3.4.7 Passive buzzer

The Haoyue development board is equipped with a passive buzzer on the backplane, which requires an input PWM wave to drive the buzzer to sound. Compared with the active buzzer, it has the advantage of controllable sound frequency, while the active buzzer can only emit a single tone. In the development board, it can be used as a sound device for information prompts.

The schematic diagram of the passive buzzer is shown in 3.4-14:

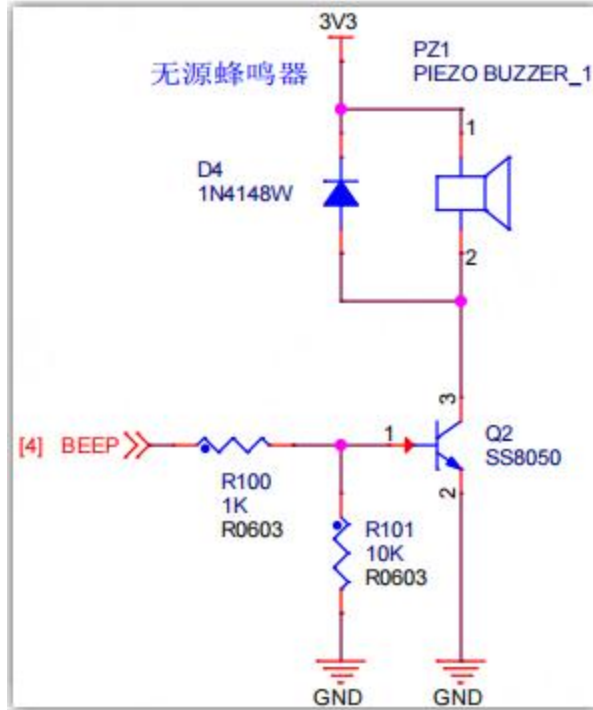


Figure 3.4-14 Passive buzzer schematic diagram

The actual picture of the passive buzzer on the development board is shown in Figure 3.4-15:



Figure 3.4-15 Actual picture of passive buzzer

The passive buzzer pin assignments are shown in the table below:

引脚名	FPGA 绑定引脚
BEEP	U14

3.4.8 USB to serial port

The development board has an onboard USB interface and is equipped with a CH340G USB to serial port chip. This interface can be used for debugging and verification of the RS232 serial port. The interface is connected to the PS end and does not require pin allocation like the FPGA. The schematic diagram is shown in Figure 3.4-16:

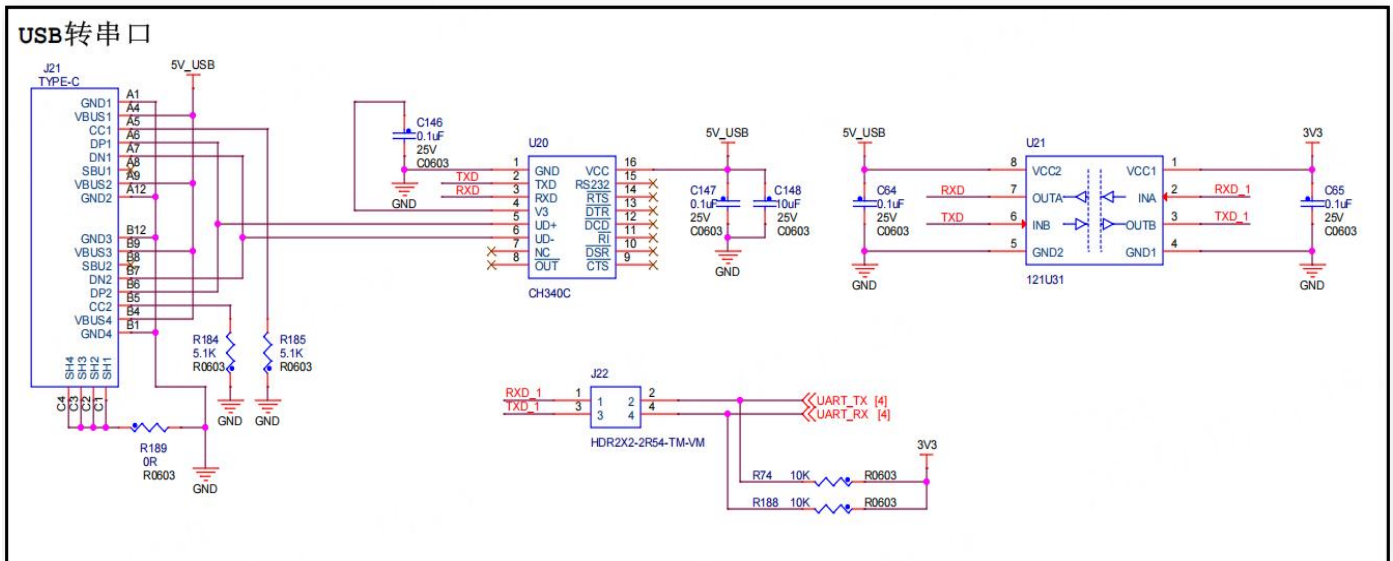


Figure 3.4-16 USB to serial port schematic diagram

The physical diagram of the USB to serial port part of the Haoyue development board is shown in Figure 3.4-17.

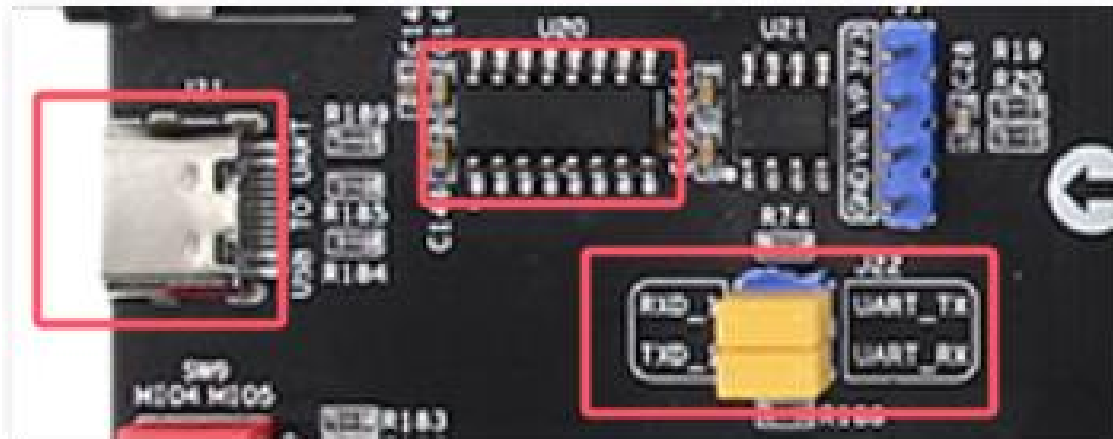


Figure 3.4-17 USB to serial port physical diagram

3.4.9 Buttons

There are multiple buttons on the bottom of the Haoyue development board for users to use. The first is a PS-side reconfiguration mechanical button. This button can be used for PS-side reconfiguration. That is to say, when we download the program into the PS side of the zynq chip, we press this button to clear the program downloaded on the PS side. The schematic diagram is shown in Figure 3.4-18:

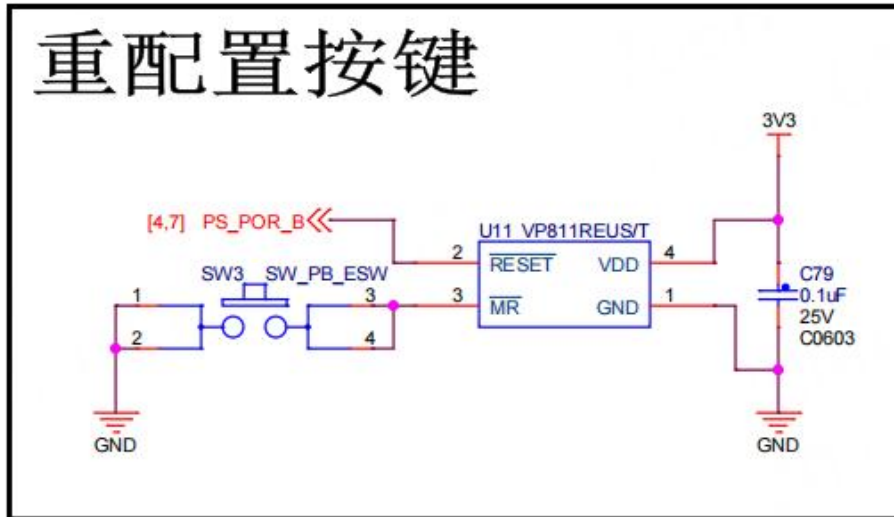


Figure 3.4-18 PS side reconfiguration button schematic diagram

The physical diagram of the PS side reconfiguration button on the Haoyue development board is shown in Figure 3.4-19:

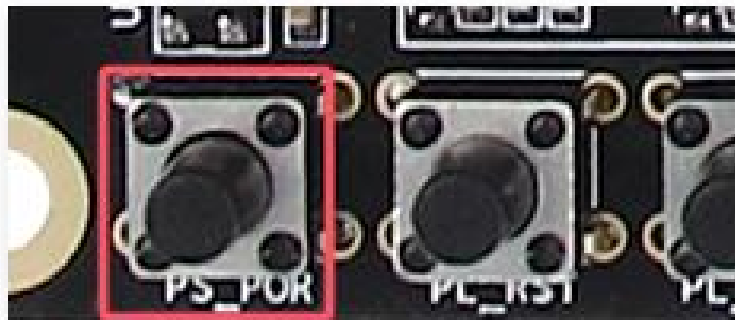


Figure 3.4-19 Physical picture of the PS terminal reconfiguration button

The PS side of the Haoyue development board baseboard is also equipped with two user-controlled mechanical buttons for controlling signal input. The schematic diagram is shown in Figure 3.4-20:

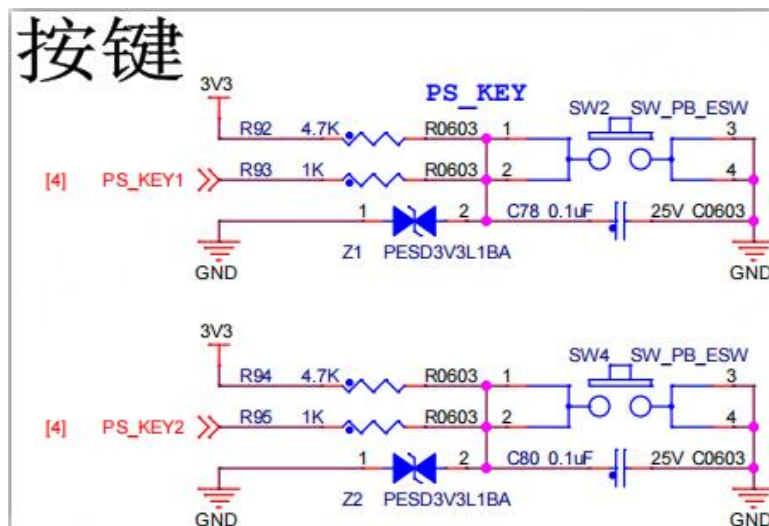


Figure 3.4-20 PS side button schematic diagram

The physical picture of the PS side buttons on the Haoyue development board is shown in Figure 3.4-21:

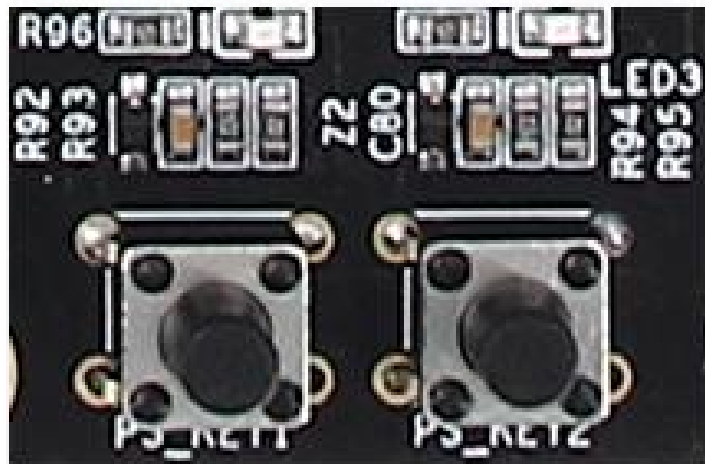


Figure 3.4-21 Actual picture of PS side buttons

The PL side of the Haoyue development board backplane is also equipped with a mechanical button for program reset and three user-controlled mechanical buttons for program reset and control signal input. The schematic diagram is shown in Figure 3.4-22:

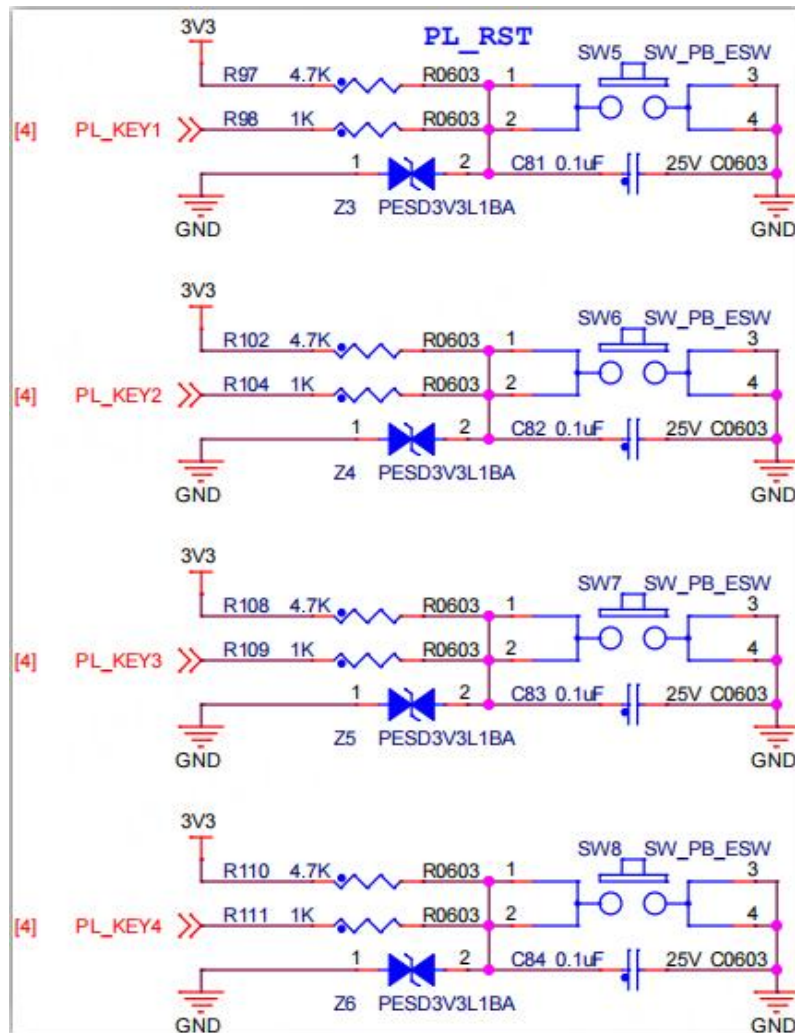


Figure 3.4-22 PL side button schematic diagram

The physical diagram of the PL side buttons on the Haoyue development board is shown in Figure 3.4-23:



Figure 3.4-23 Actual picture of PL side buttons

The PL side button pin assignment is as shown in the following table:

引脚名	FPGA 绑定引脚
PL_KEY1	G19
PL_KEY2	G20
PL_KEY3	H15
PL_KEY4	G15

3.4.10 LED lights

The Haoyue development board has 7 LED user indicators on the backplane, one of which is the power indicator, 2 user indicators on the PS side, and 4 user indicators on the PL side. The first is the power light. When the development is powered on, the power indicator light will light up. The schematic diagram is shown in Figure 3.4-294:

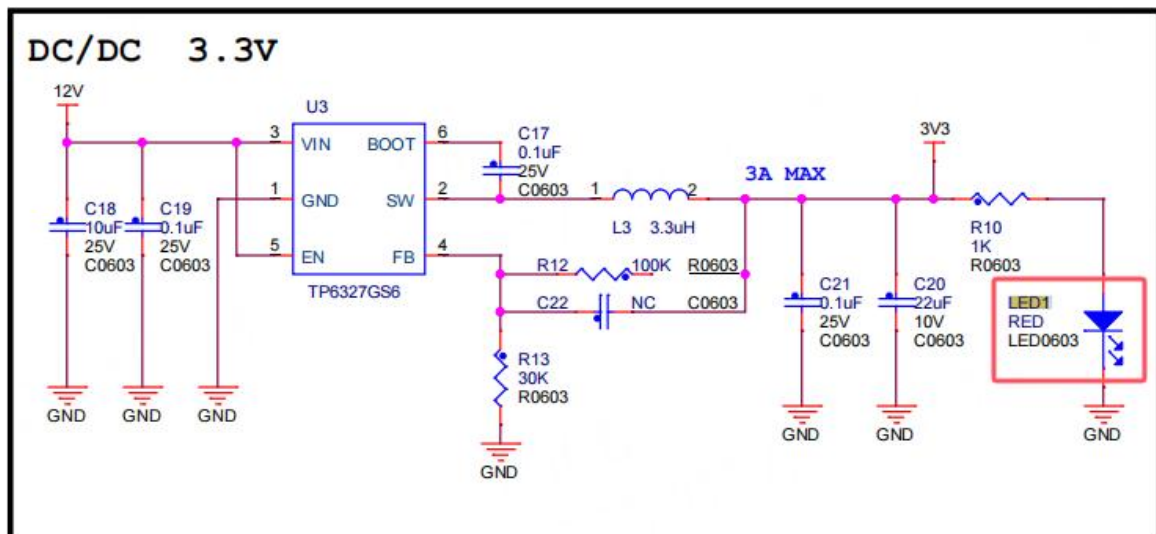


Figure 3.4-24 Schematic diagram of power indicator light

Secondly, there are two LED user indicators on the PS side, which can be used as program indication signals. The schematic

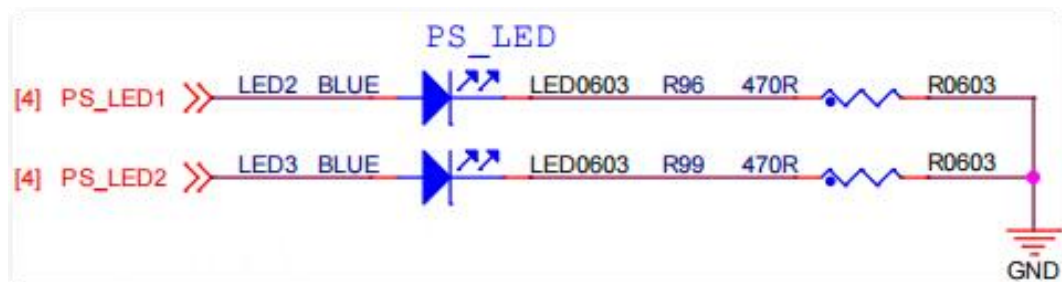


Figure 3.4-25 PS end user LED schematic diagram

Finally, there are 4 LED user indicators on the PL side, which can be used as program indication signals. The schematic diagram is as follows:

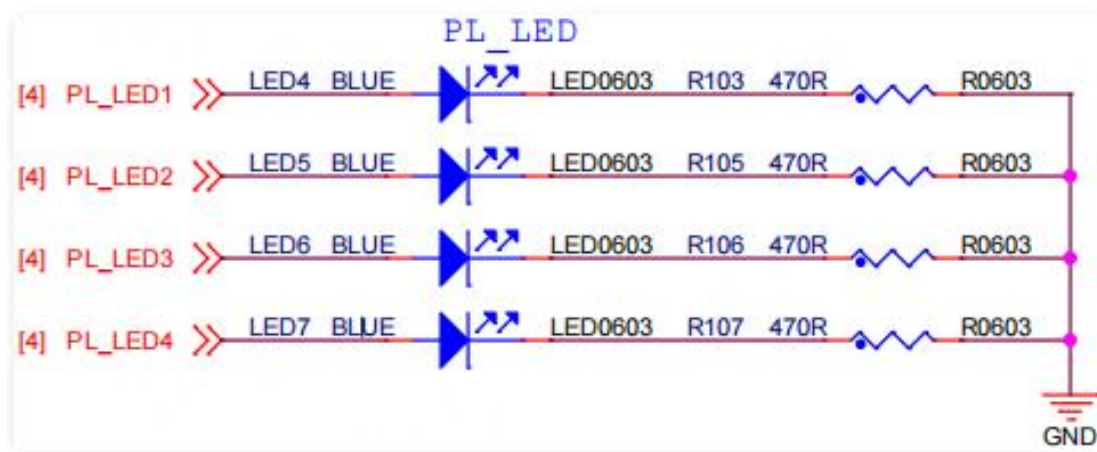


Figure 3.4-26 PL end user LED schematic diagram

Figure 3.4-27 shows the actual picture of the LED lights on the development board. The ① box is the power indicator light; the ② box is 2 LED lights located on the PS side for users to use; the ③ box is 4 LED lights located on the PL side for users to use.

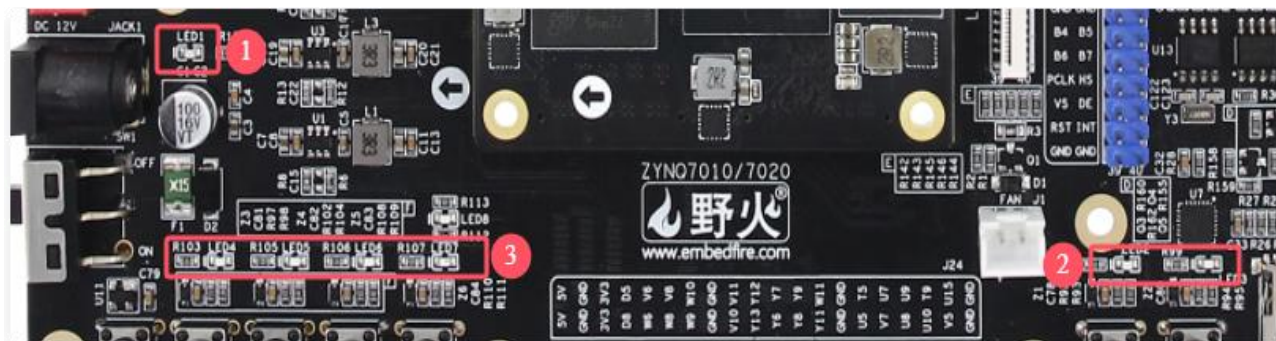


Figure 3.4-27 Actual picture of the LED light on the backplane of Haoyue development board

The pin assignment of the PL side LED light is as shown in the following table:

引脚名	FPGA 绑定引脚
PL_LED1	K16
PL_LED2	J16
PL_LED3	K14
PL_LED4	J14

3.4.11 TFT_LCD LCD screen interface

There is an onboard TFT_LCD LCD screen interface, which can be connected to the Yehuo TFT_LCD LCD screen for picture display, and supports RGB565/888 screens. The schematic diagram is shown in Figure 3.4-28. It can be seen from the figure that the RGB888 image mode is used, the bit width is 24bit, the high 8 bits represent blue, the low 8 bits represent red, and the middle 8 bits represent green. RGB888 can display more types of colors than an RGB565, and the displayed image is more delicate.

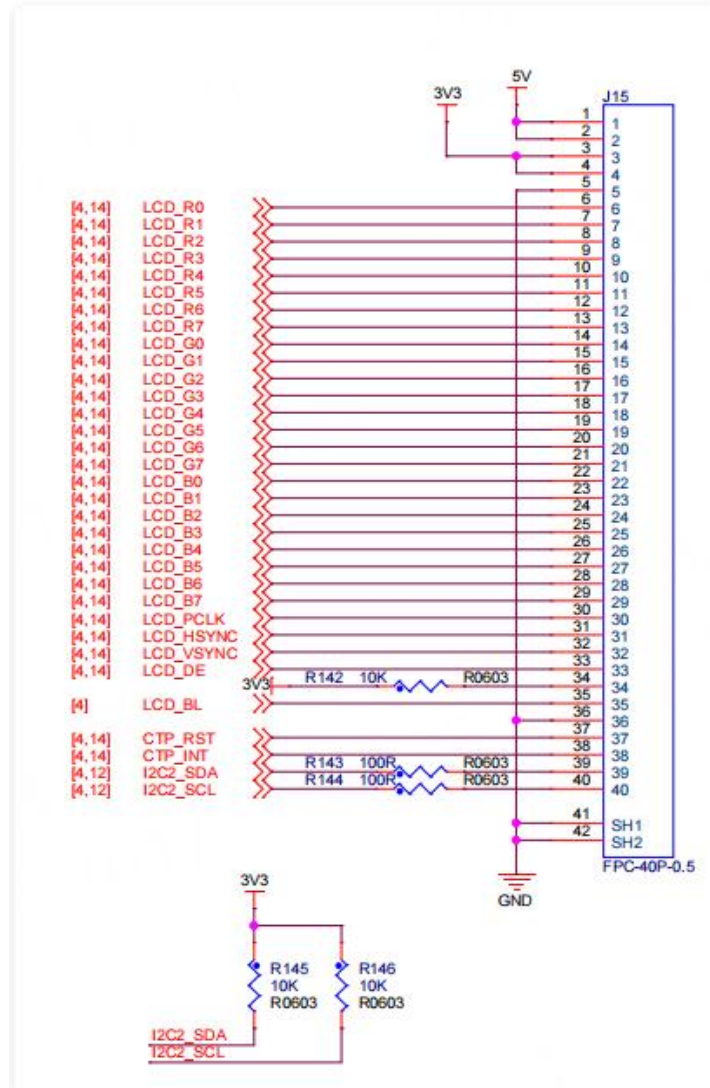


Figure 3.4-28 TFT_LCD interface schematic diagram

Figure 3.4-29 shows the physical diagram of the TFT_LCD interface.



Figure 3.4-29 TFT_LCD interface physical diagram

The connection method of YeFire TFT_LCD screen is shown in Figure 3.4-30:



引脚名	FPGA 绑定引脚
LCD_BL	T19
CTP_RST	R19
CTP_INT	P19
I2C2_SDA	N20
I2C2_SCL	P20
LCD_VSYNC	W13
LCD_HSYNC	V12
LCD_DE	V15
LCD_PCLK	W15
LCD_R0	T11
LCD_R1	T10
LCD_R2	T12
LCD_R3	U12
LCD_R4	U13
LCD_R5	V13
LCD_R6	T14
LCD_R7	T15
LCD_G0	V17
LCD_G1	V18
LCD_G2	T16
LCD_G3	U17
LCD_G4	P15
LCD_G5	P16
LCD_G6	P14
LCD_G7	R14
LCD_B0	V16
LCD_B1	W16
LCD_B2	R16
LCD_B3	R17
LCD_B4	W18
LCD_B5	W19
LCD_B6	U18
LCD_B7	U19

3.4.12 RS232

The development board has an onboard RS-232 interface and is equipped with a MAX3232 transceiver chip, which can convert the "TTL" level into the "RS232" level standard for serial communication between the development board and external devices. The schematic diagram is shown in Figure 3.4-31:

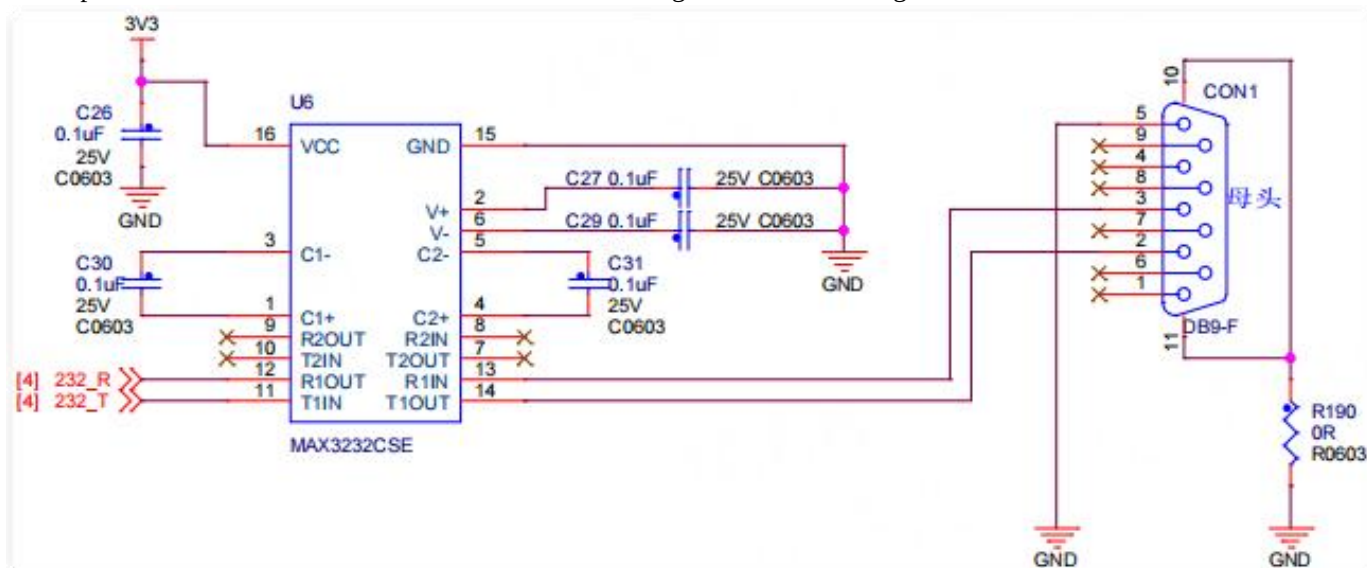


Figure 3.4-31 RS232 serial port schematic diagram

The physical diagram of the serial port interface and transceiver chip on the Haoyue development board is shown in Figure



Figure 3.4-32 Physical picture of RS232 on Haoyue development board

The RS232 pin assignment is shown in the table below:

引脚名	FPGA 绑定引脚
232_R	B20
232_T	C20

3.4.13 RS485

There is an onboard RS485 bus interface and the transceiver is the MAX3485 chip. Communicate with external RS485 devices through the two ports 485A and 485B. When communicating, you need to connect port A to port A and port B to port B for normal communication. The schematic diagram is shown in Figure 3.4-33:

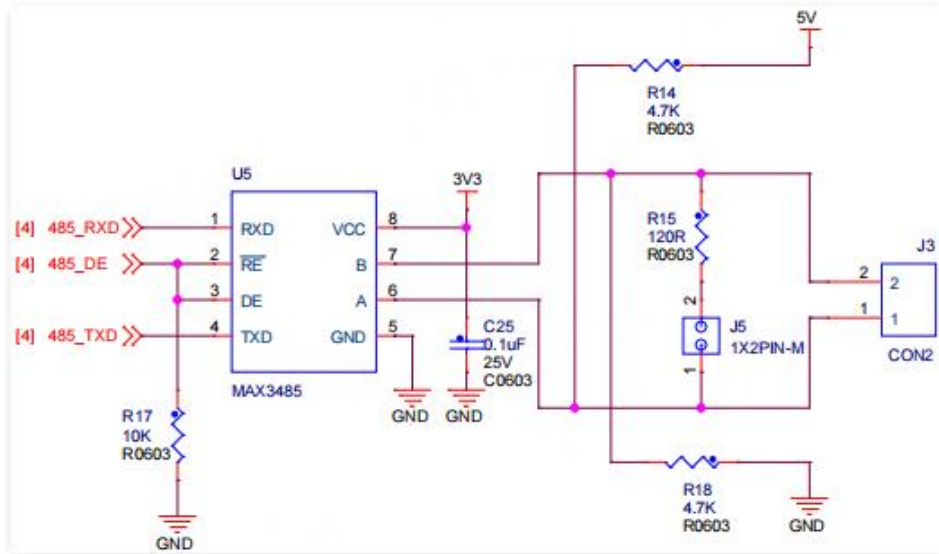


Figure 3.4-33 Onboard RS485 interface schematic diagram

The RS485 interface on the Haoyue development board and the physical diagram of the MAX3485 chip are shown in Figure 3.4-34:



Figure 3.4-34 Physical diagram of onboard RS485 interface

The RS485 pin assignment is shown in the table below:

引脚名	FPGA 绑定引脚
485_RXD	D20
485_TXD	D19
485_DE	G14

3.4.14 CAN interface

There is an onboard CAN communication bus interface, and the transceiver is the TJA1042 chip. Communicate with external CAN devices through CANH and CANL ports. For the same communication, H port must be connected to H port and L port to L port for normal communication. The schematic diagram is shown in 3.4-35:

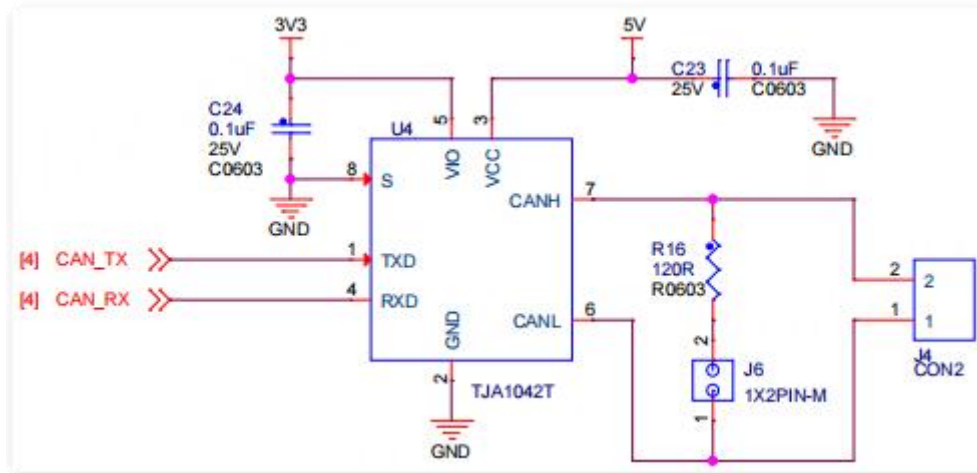


Figure 3.4-35 CAN interface schematic diagram

The CAN interface and the physical picture of the TJA1042 chip are shown in Figure 3.4-36

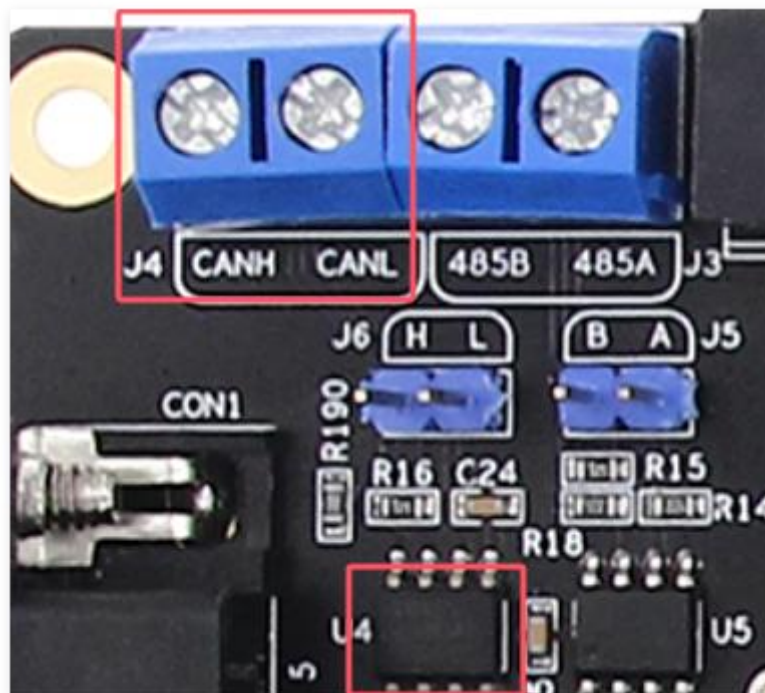


Figure 3.4-36 Physical diagram of CAN interface

The CAN pin assignments are shown in the table below:

引脚名	FPGA 绑定引脚
CAN_TX	B19
CAN_RX	A20

3.4.15 Real-time clock

The development board is equipped with a real-time clock RTC chip, model number PCF8563. PCF8563 is an industrial-grade ultra-low-power multi-function clock/calendar chip launched by Philips that includes an I2C bus interface function. PCF8563 has a variety of alarm functions, timer functions, clock output functions and interrupt output functions, and can complete various complex timing services. The schematic diagram of the real-time clock is shown in Figure 2.4-38. It needs to be connected to an external passive clock of 32.768KHz to provide an accurate clock source for the clock chip. At the same time, in order to ensure that the clock can still run normally after power failure, the development board is also equipped with a button battery interface to supply power to the clock chip. In this way, if we connect the button battery, the real-time clock can still run normally even if the development board loses power.

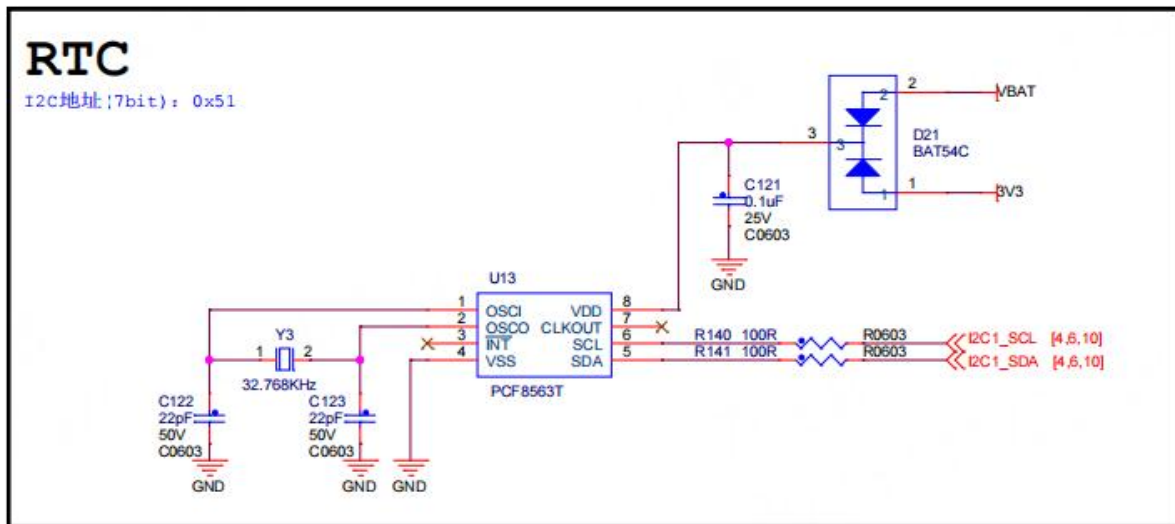


Figure 3.4-37 Real-time clock schematic diagram

The physical diagram of the real-time clock chip and RTC base (model CR1220) on the development board is shown in Figure 3.4-38. The RTC base is used to place a button battery to power the real-time clock chip.

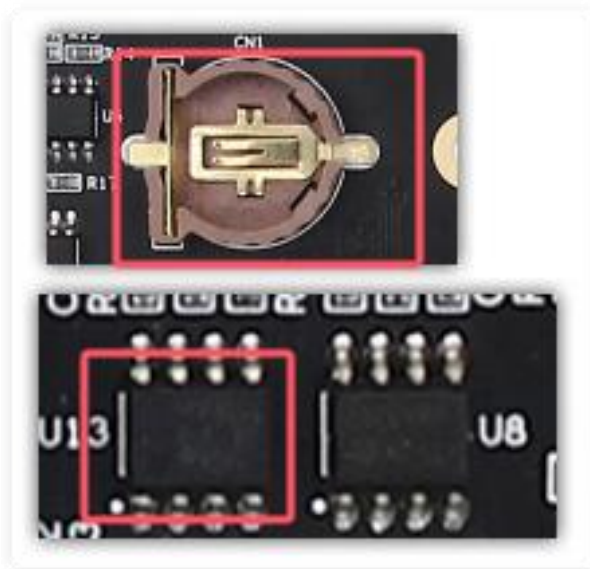


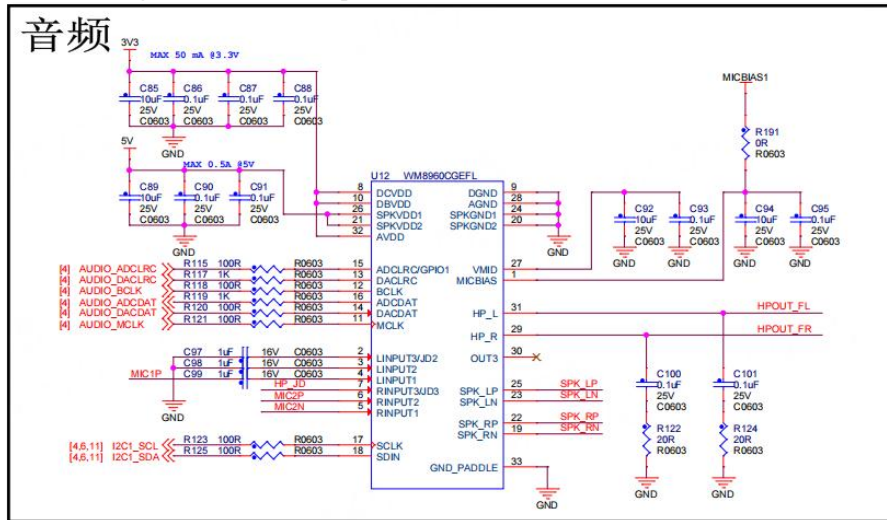
Figure 3.4-38 Real-time clock physical diagram

The real-time clock pin assignments are shown in the table below:

引脚名	FPGA 绑定引脚
I2C1_SCL	N17
I2C1_SDA	P18

3.4.16 Audio

There is an audio chip on the board, model WM8960. This audio chip can be used to encode and decode audio data, so it can realize audio loopback playback, recording and other functions. As shown in Figure 3.4-39, this figure is the main schematic diagram of the audio part.



3.4.17 BOOT switch

There is a BOOT mode selection switch onboard, which can adjust the startup mode of the board. The Haoyu e development board has three startup modes, namely FLASH, JTAG, and SD card. The schematic diagram is as follows:

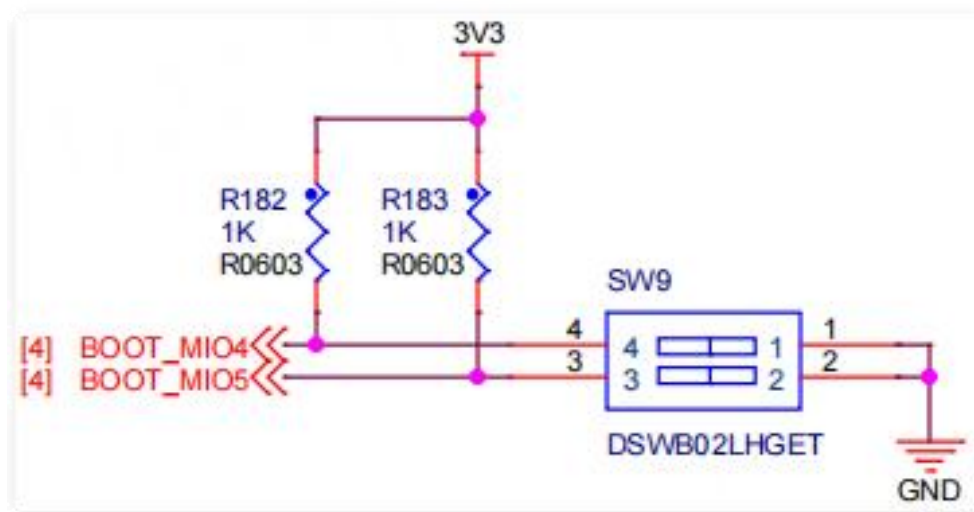


Figure 3.4-41 BOOT switch schematic diagram

The physical diagram of the BOOT switch on the development board is shown in Figure 3.4-42.



Figure 3.4-42 Physical picture of BOOT switch

BOOT switch mode selection is shown in the following table:

BOOT_MODE	JTAG	NAND	QSPI_FLASH	SD_CARD
MIO4	0	1	0	1
MIO5	0	0	1	1

3.4.18 Camera interface

The development board is equipped with a 20-pin DVP camera interface. This interface can be connected to the Wildfire OV7725 camera or OV5640 camera (the camera module needs to be purchased) to collect real-time images. After collection, it can be output to a VGA display, TFT LCD screen, or HDMI screen for display through the VGA interface, LCD interface, or HDMI interface on the development board. Among them, OV7725 supports the output of images up to 300,000 pixels (640 x 480 resolution), and OV5640 supports the output of images up to 5 million pixels (2592 x 1944 resolution). You can choose according to your needs.

The schematic diagram of the camera interface is shown in Figure 3.4-43:

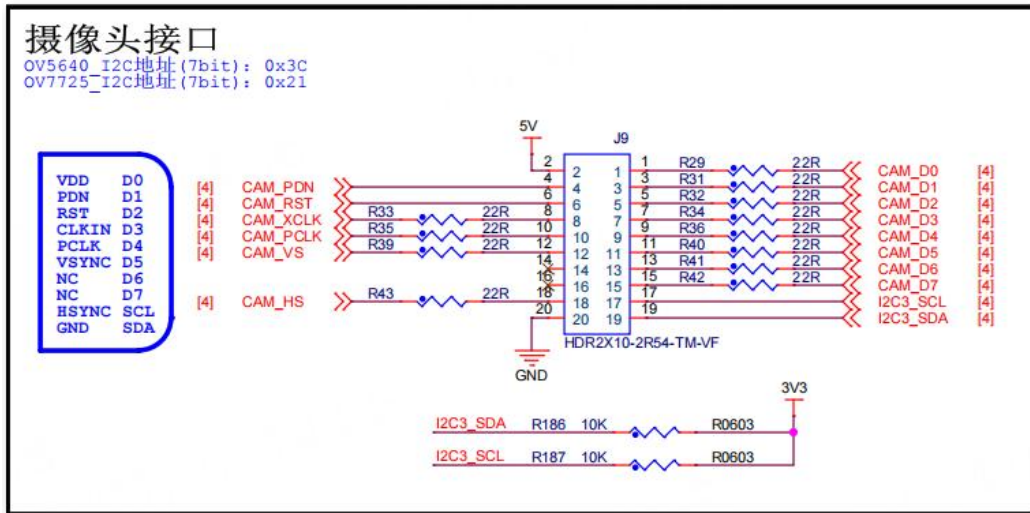


Figure 3.4-43 Camera interface schematic diagram

The physical diagram of the development board camera interface is shown in Figure 3.4-44:

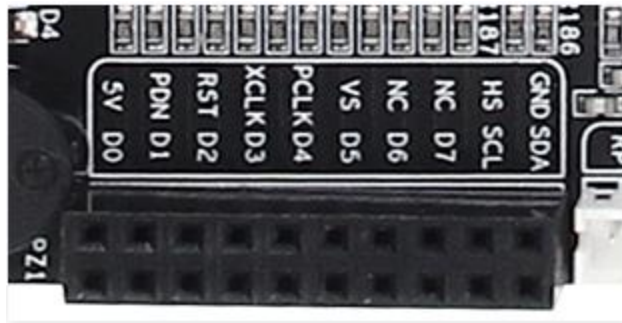


Figure 3.4-44 Actual picture of camera interface

This interface can be connected to the Wildfire OV7725 camera or OV5640 camera. The connection method is as shown in



Figure 3.4-45 OV5640 camera connected to development board

The camera pin assignment is shown in the table below:

引脚名	FPGA 绑定引脚
CAM_PDN	M17
CAM_RST	M18
CAM_XCLK	L20
CAM_PCLK	L19
CAM_VS	M19
CAM_HS	M20
CAM_D0	N15
CAM_D1	N16
CAM_D2	L14
CAM_D3	L15
CAM_D4	M14
CAM_D5	M15
CAM_D6	L16
CAM_D7	L17
I2C3_SCL	T17
I2C3_SDA	R18

3.4.19 USB2.0

The Haoyue development board is equipped with four USB HOST interfaces. In addition, there is also a USB SLAVE interface. The USB SLAVE defaults to the host mode. When the slave USB interface is connected to the P C, it automatically switches to the slave mode. In addition, it should be noted that USB SLAVE and USB HOST cannot be used at the same time. The schematic diagram of this part is as follows:

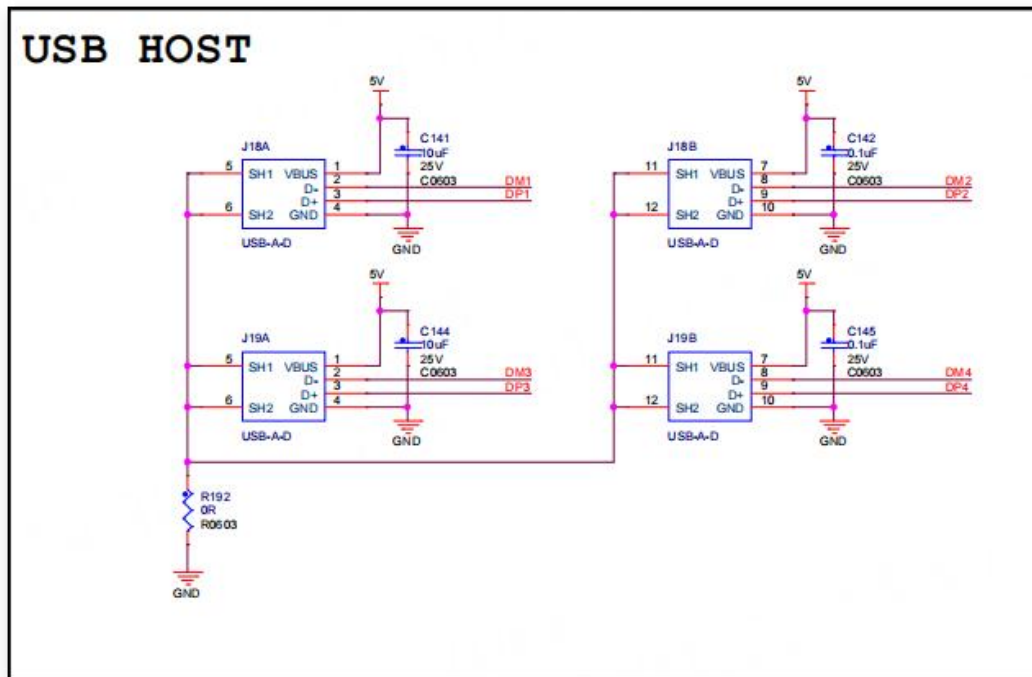


Figure 3.4-46 USB HOST schematic diagram

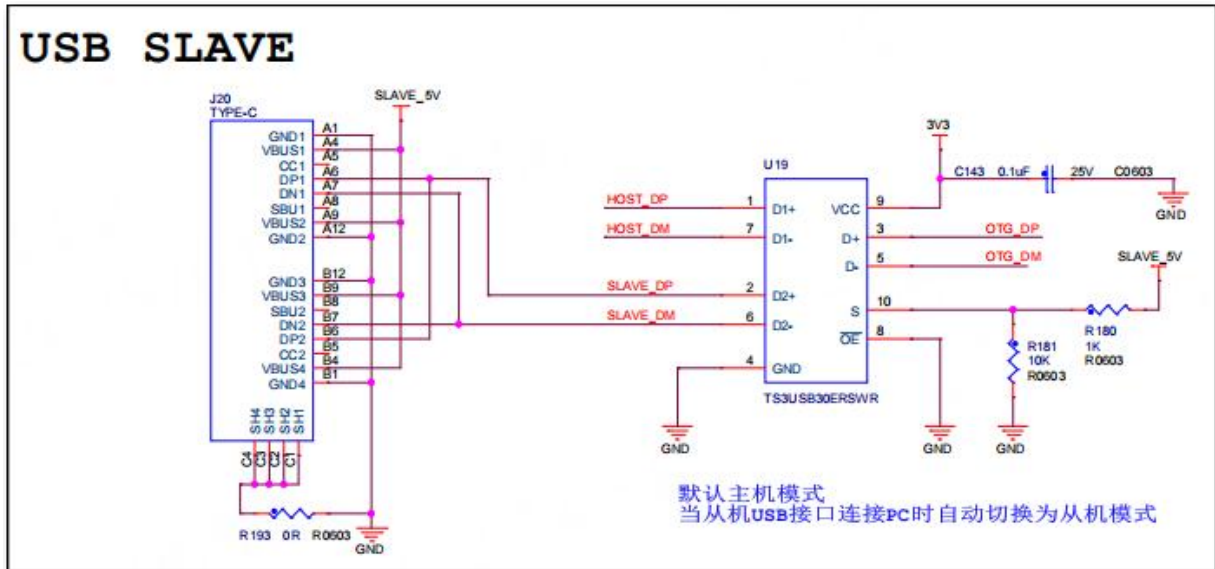


Figure 3.4-47 USB SLAVE schematic diagram

Figure 3.4-48 shows the physical diagram of the USB interface on the development board. Among them, ① is the USB HOST



Figure 3.4-48 Physical picture of USB interface

3.4.20 Expand IO

The development board is equipped with rich expansion IO port resources. The two 40P IO ports can meet most expansion module designs. It can also be connected to the Wildfire AD/DA module (optional) for analog-to-digital conversion. It should be noted that J24 can only be used by zynq7020.

The schematic diagram of the development board expansion IO port is shown in Figure 3.4-49:

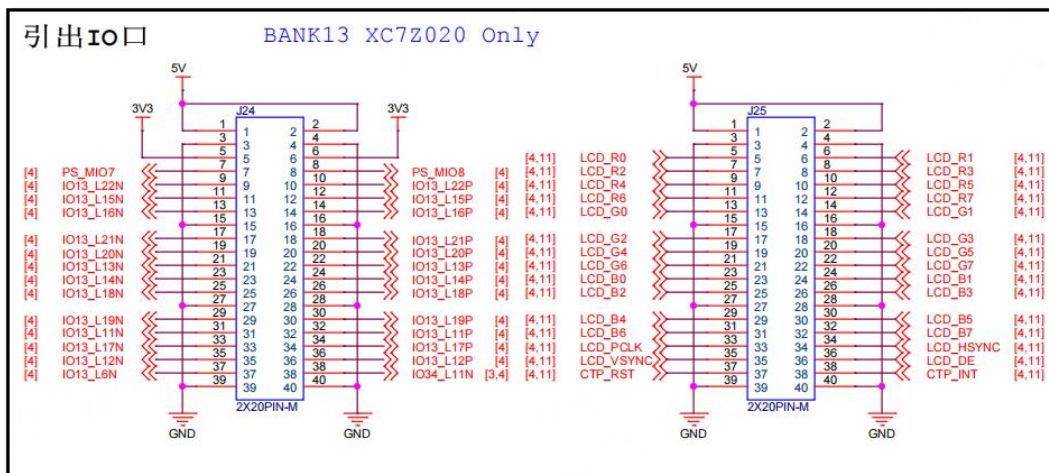


Figure 3.4-49 Extended IO schematic diagram

The physical diagram of the expansion IO port on the development board is shown in Figure 3.4-50:

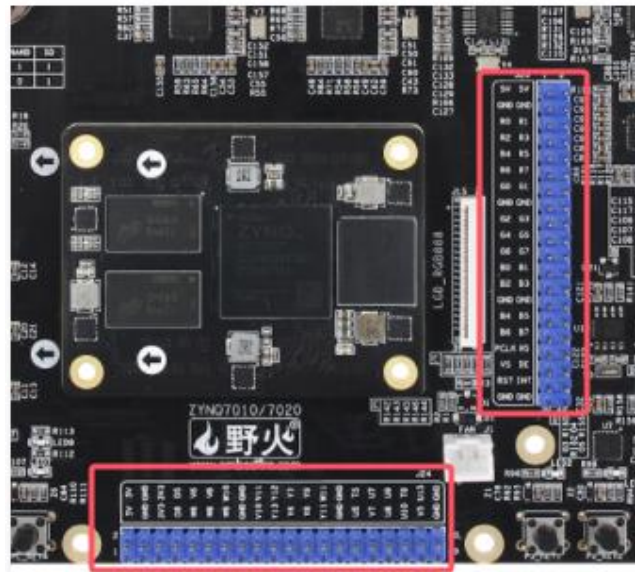


Figure 3.4-50 Physical diagram of expanded IO

The expansion IO can be connected to the Yehuo high-speed AD/DA module and OV5640 binocular camera. The connection

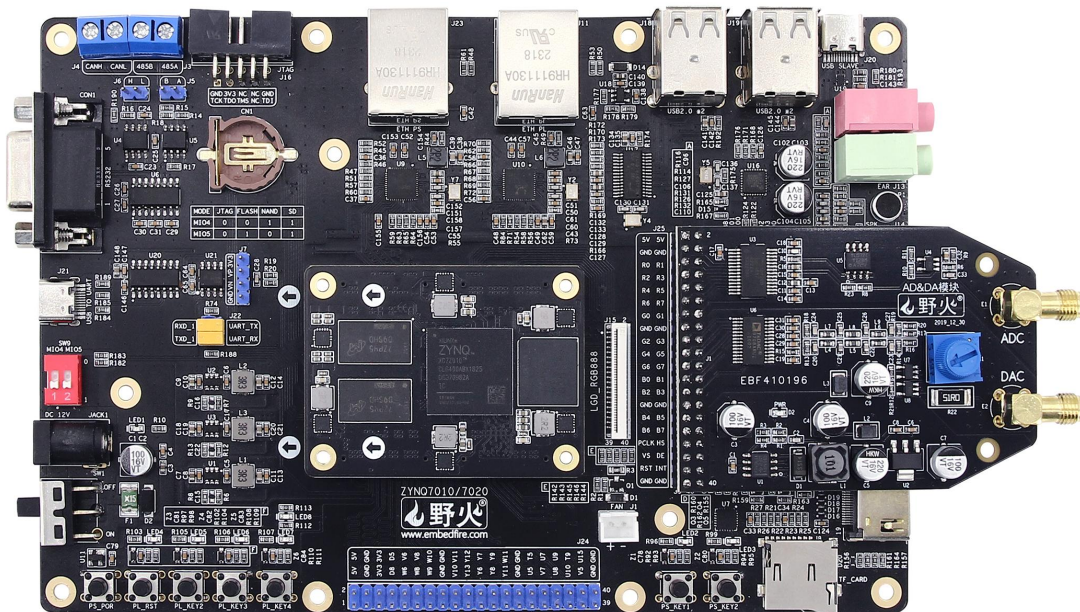


Figure 3.4-51 Wildfire high-speed AD/DA module and development board connection

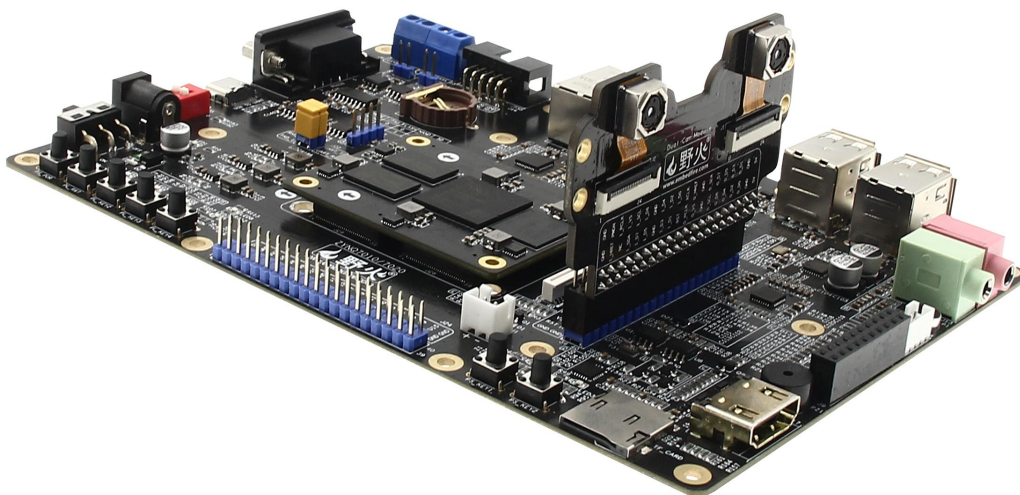


Figure 3.4-52 Wildfire binocular camera module and development board connection

The J24 expansion pin assignment is shown in the following table:

引脚编号	FPGA 绑定引脚	引脚编号	FPGA 绑定引脚
1	5V	2	5V
3	GND	4	GND
5	3V3	6	3V3
7	D8	8	D5
9	W6	10	V6
11	W8	12	V8
13	W9	14	W10
15	GND	16	GND
17	V10	18	V11
19	Y13	20	Y12
21	Y6	22	Y7
23	Y8	24	Y9
25	Y11	26	W11
27	GND	28	GND
29	U5	30	T5
31	V7	32	U7
33	U8	34	U9
35	U10	36	T9
37	V5	38	U15
39	GND	40	GND

The J25 expansion pin assignment is shown in the following table:

引脚编号	FPGA 绑定引脚	引脚编号	FPGA 绑定引脚
1	5V	2	5V
3	GND	4	GND
5	T11	6	T10
7	T12	8	U12
9	U13	10	V13
11	T14	12	T15

13	V17	14	V18
15	GND	16	GND
17	T16	18	U17
19	P15	20	P16
21	P14	22	R14
23	V16	24	W16
25	R16	26	R17
27	GND	28	GND
29	W18	30	W19
31	U18	32	U19
33	W15	34	V12
35	W13	36	V15
37	R19	38	P19
39	GND	40	GND